Teaching Portfolio

HANG ZHANG

Digital Building Technologies · ETH Zurich

Research-led teaching in computational design, fabrication, and collaborative prototyping.

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Teaching Tutor

Select a project title to jump to its opening slide.

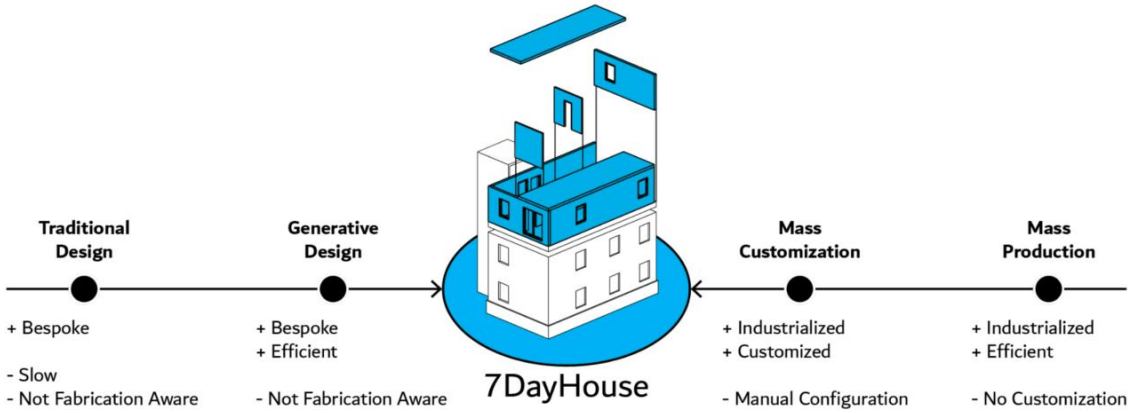
DigitalFUTURES 2023 · RL-FabDesign

Reinforcement learning for fabrication-informed generative design

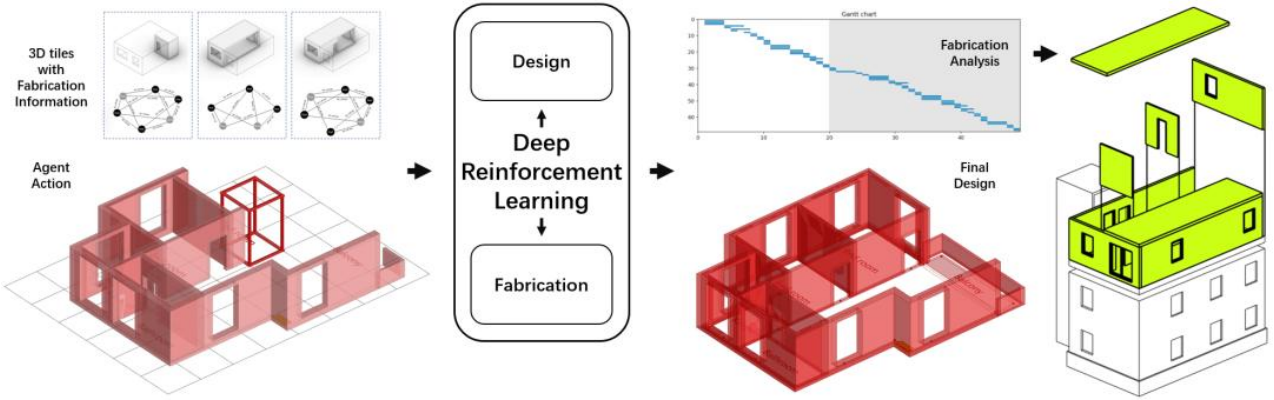
WORKSHOP HOST · 7DAYHOUSE PLATFORM

Students connect design exploration to fabrication feedback.

The workshop positioned **deep reinforcement learning** inside the 7DayHouse pipeline: design proposes spatial systems, fabrication analysis evaluates them, and the agent learns from the reward. Teaching emphasized **state/action/reward**, **fabrication-aware objectives**, interpretable iteration, and translation from **learned policy** to buildable assemblies.



Workshop framing · 7DayHouse links bespoke generation, mass customization, and industrialized production.



Teaching workflow · A DRL agent learns design moves from fabrication-analysis rewards.

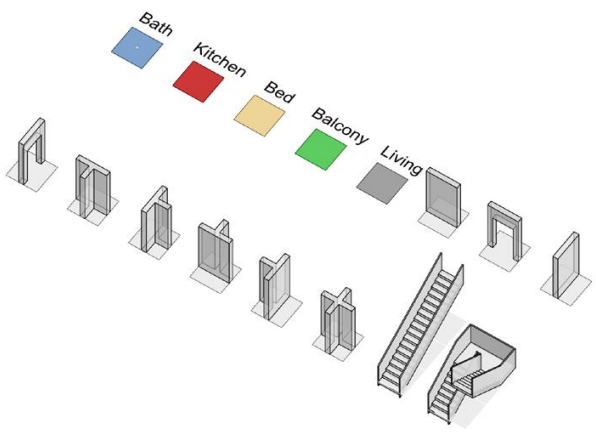
LEARNING OBJECTIVES

01 Formulate a spatial design state
02 Define fabrication-aware rewards
03 Train and read an RL policy
04 Translate outcomes into architectural proposals

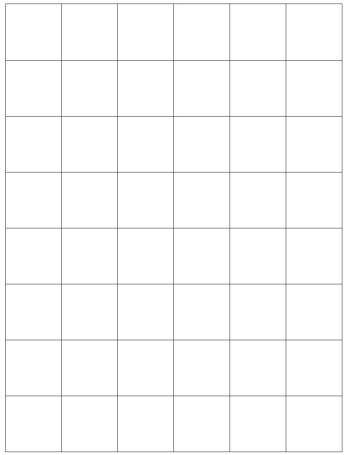
RL-FabDesign · Agent Actions

A component library supplies spatial moves for the agent

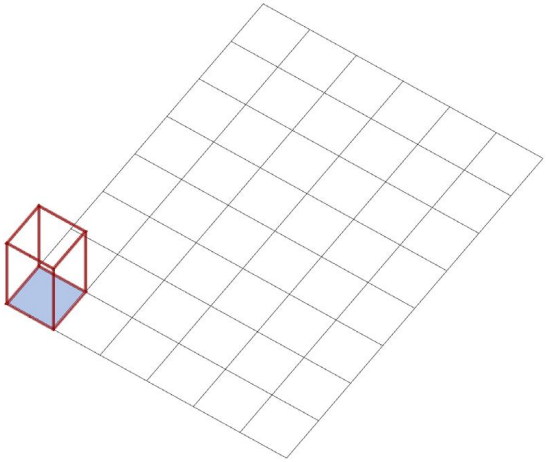
DigitalFUTURES 2023 · Day 1 teaching



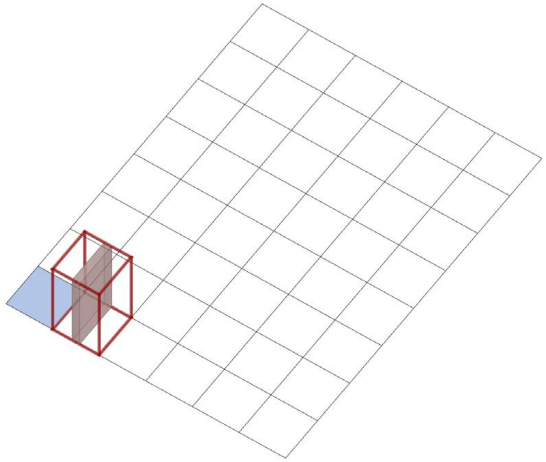
Design library



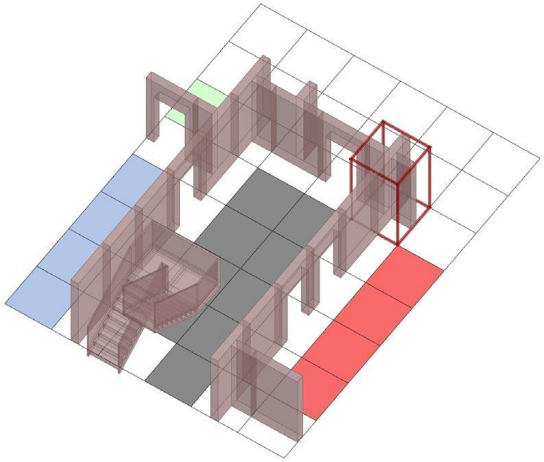
Initial grid



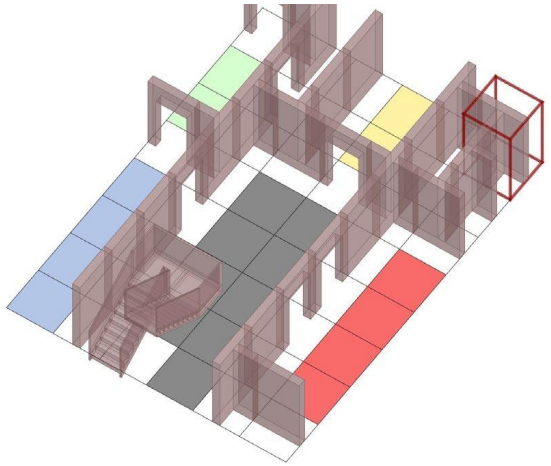
First placement



Next action



Expanded spatial state

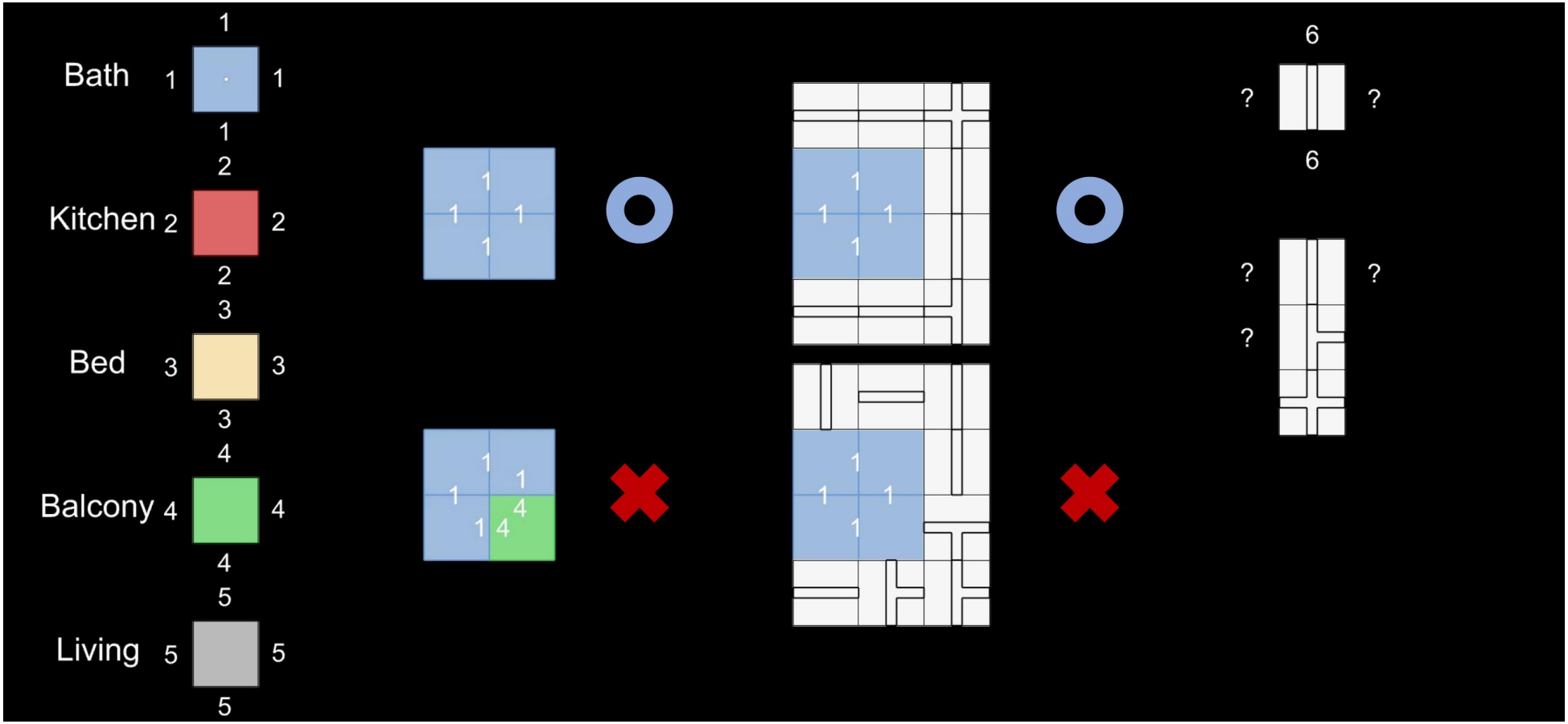


Final arrangement

RL-FabDesign · Adjacency Rules

Room relationships define which local configurations are allowed

DigitalFUTURES 2023 · Day 1 teaching

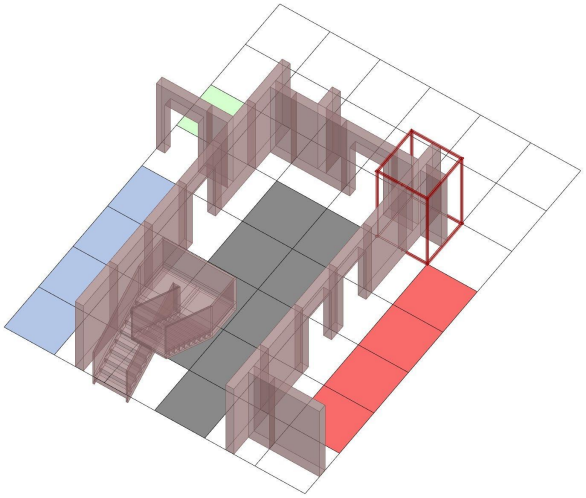


The workshop compares valid and invalid placements before an action is accepted.

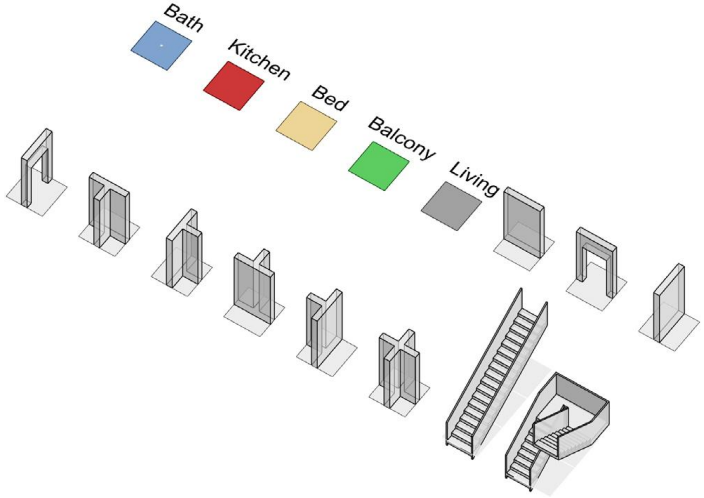
RL-FabDesign · Action Masking

Adjacency rules filter the next move from the design library

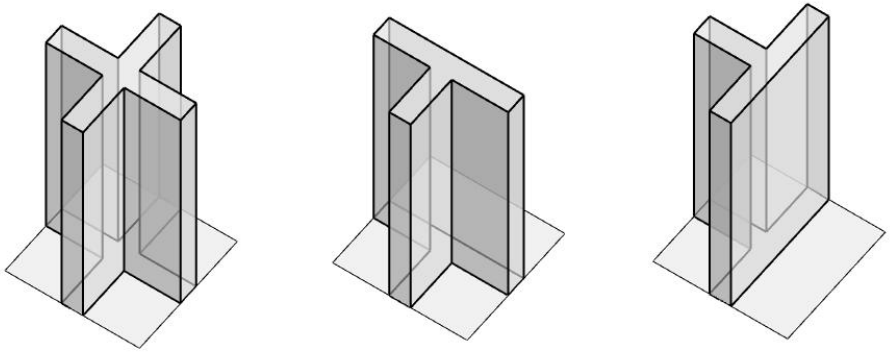
DigitalFUTURES 2023 · Day 1 teaching



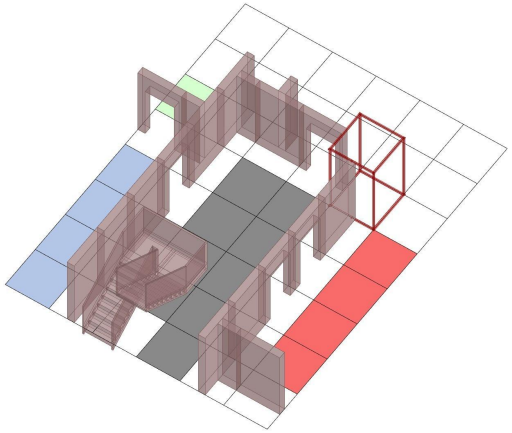
Current spatial state



Available components



Choices permitted by the rule

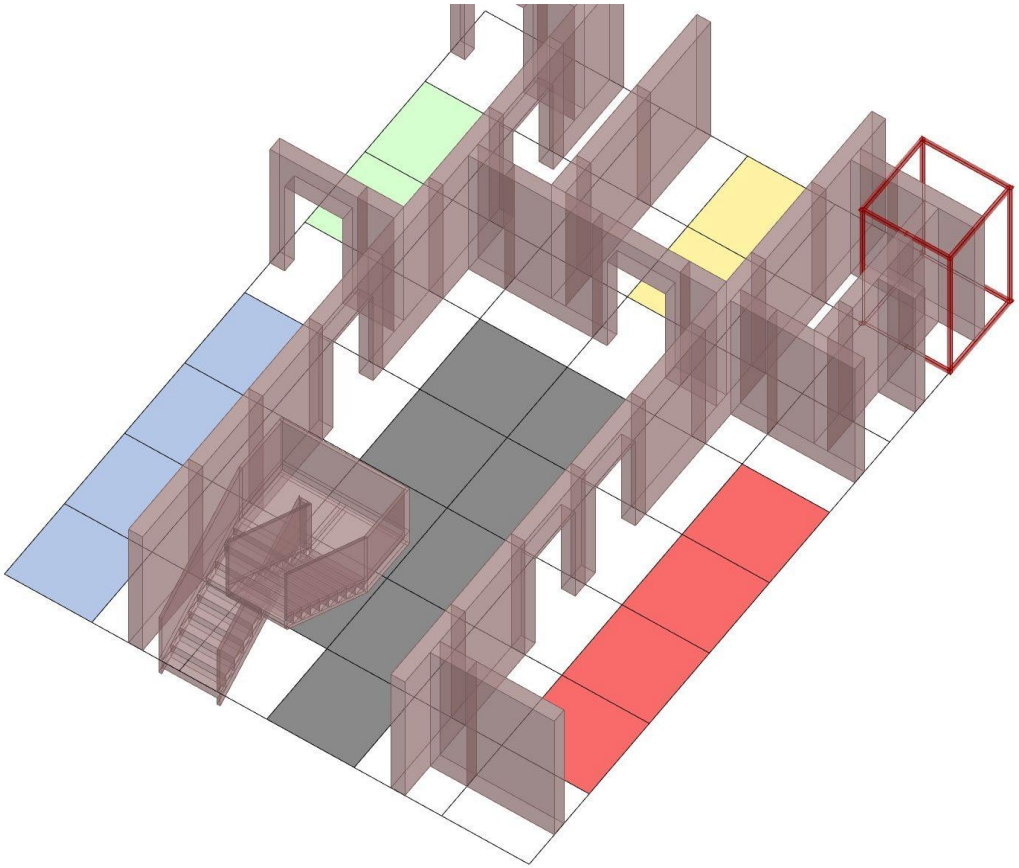


State after the selected action

RL-FabDesign · Final Reward

Students connect spatial targets to fabrication feedback

DigitalFUTURES 2023 · Day 1 teaching



Example spatial state evaluated at the end of an episode.

DESIGN TARGET

Desired room areas define the spatial target used for the final reward.

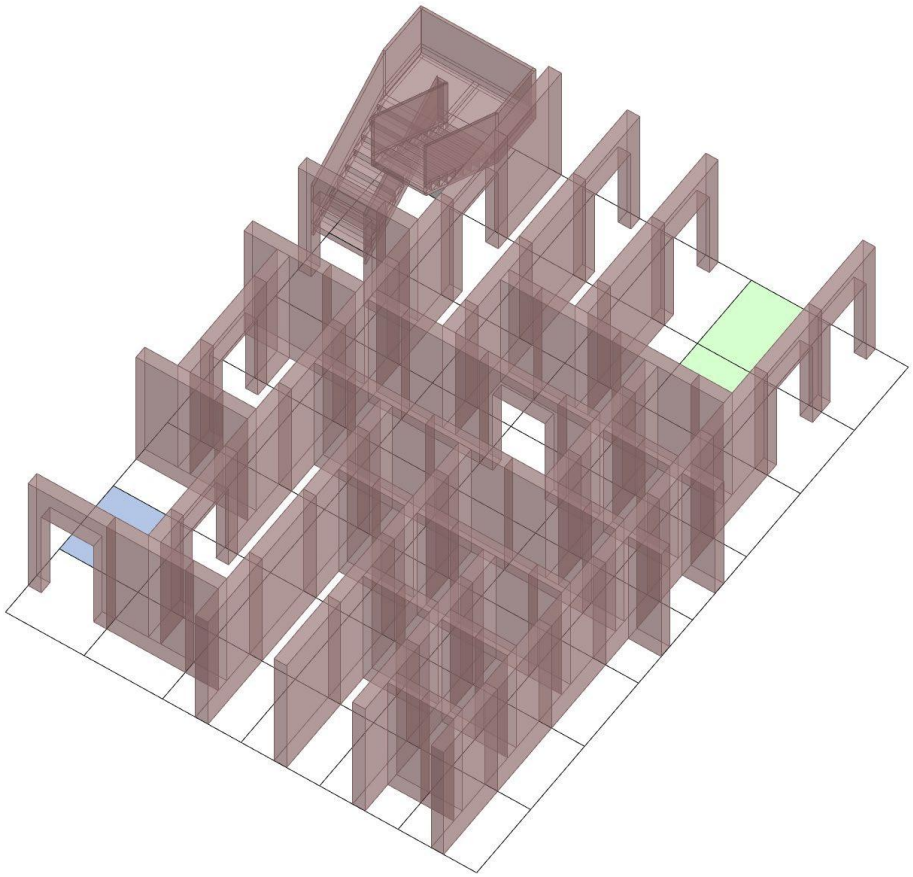
FABRICATION SIGNAL

Consumed fabrication time adds feedback about making the design.

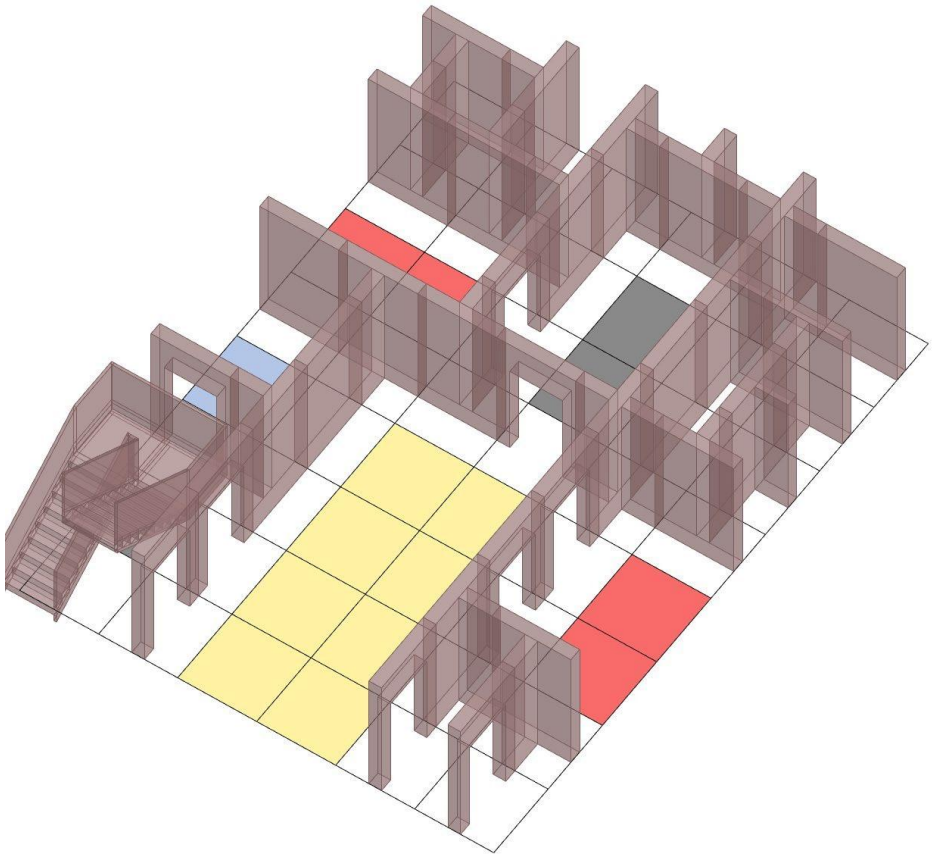
RL-FabDesign · Training Comparison

Day 1 example contrasts random generation with a trained policy

DigitalFUTURES 2023 · Day 1 teaching



Random generation · reward –275

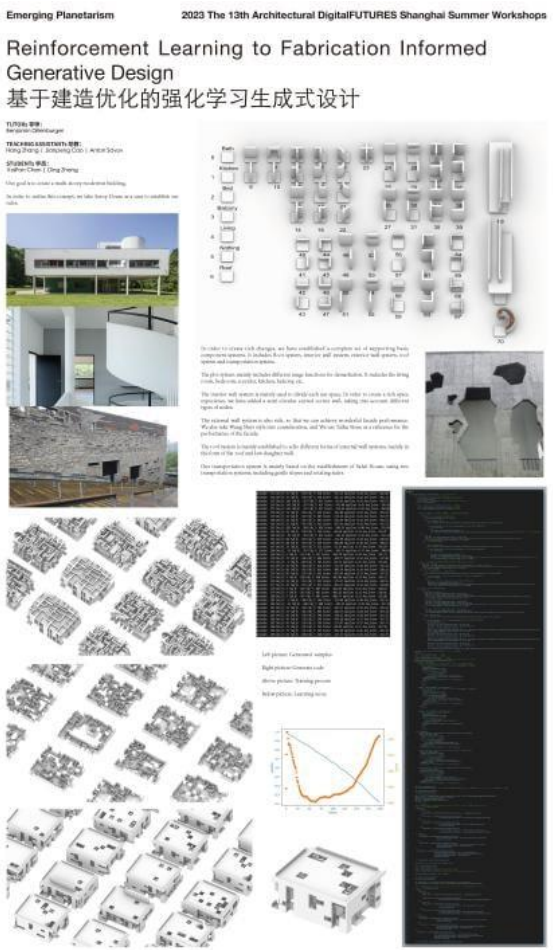


After 3,000 training episodes · reward 183

Students compare the two assemblies to discuss what the reward encouraged.

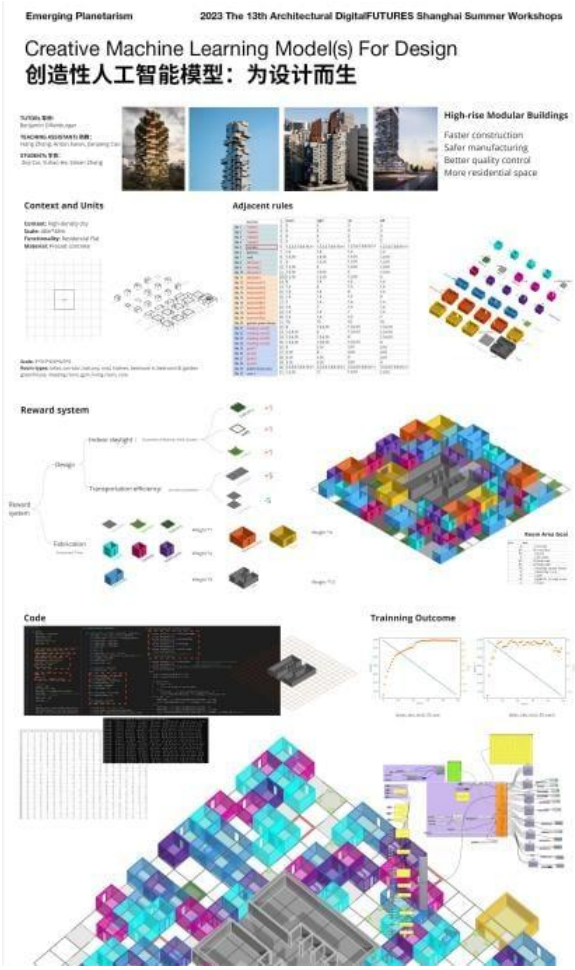
RL-FabDesign · Workshop Outcomes

Six groups, six interpretations of fabrication-informed learning



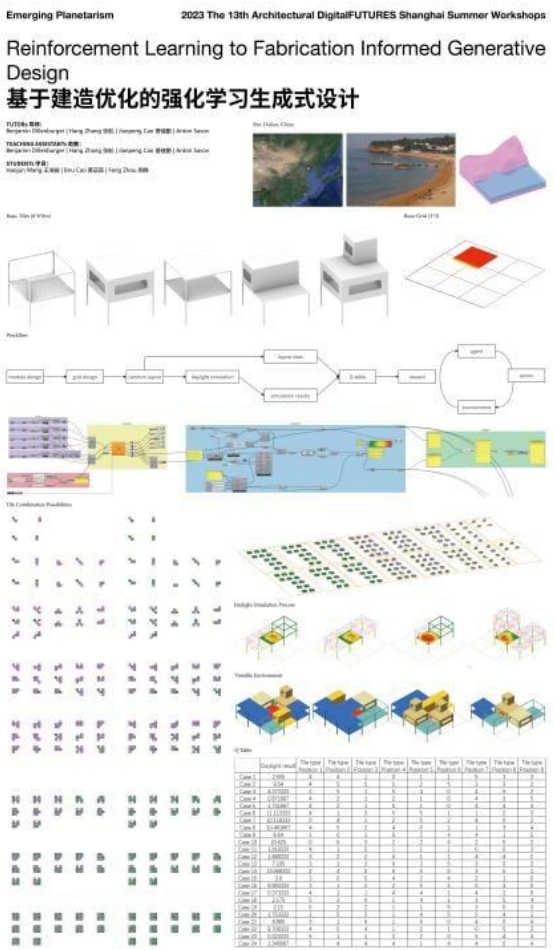
ENERGY

Rewarding environmental performance



MULTI-FLOOR

Coordinating vertical spatial systems

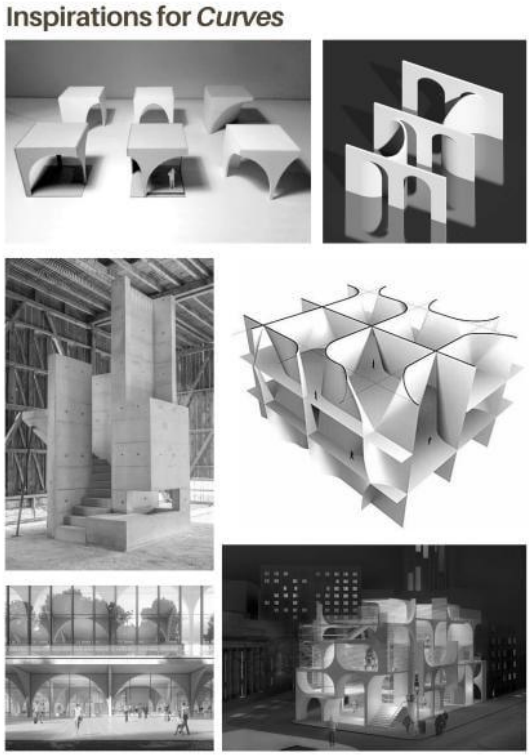


MODULE LIBRARY

Learning from repeatable components

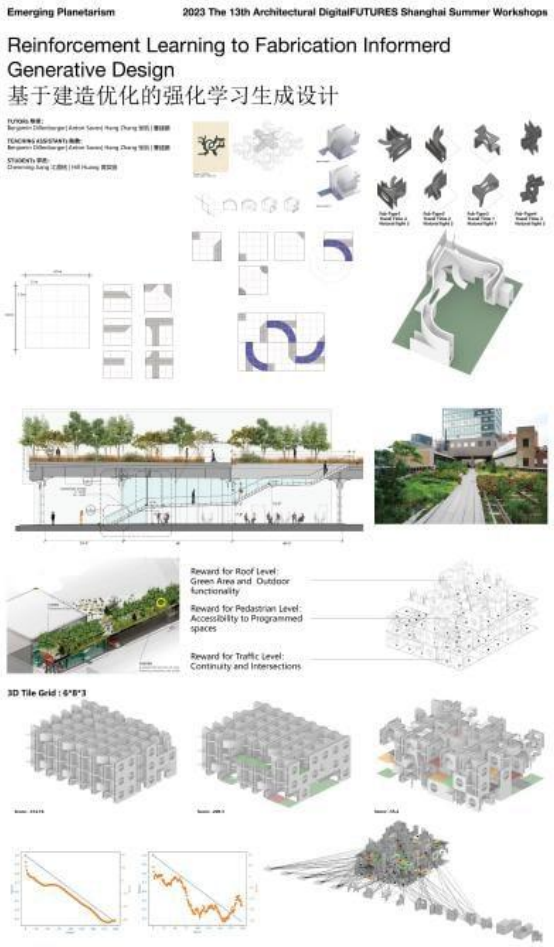
RL-FabDesign · Workshop Outcomes

Six groups, six interpretations of fabrication-informed learning



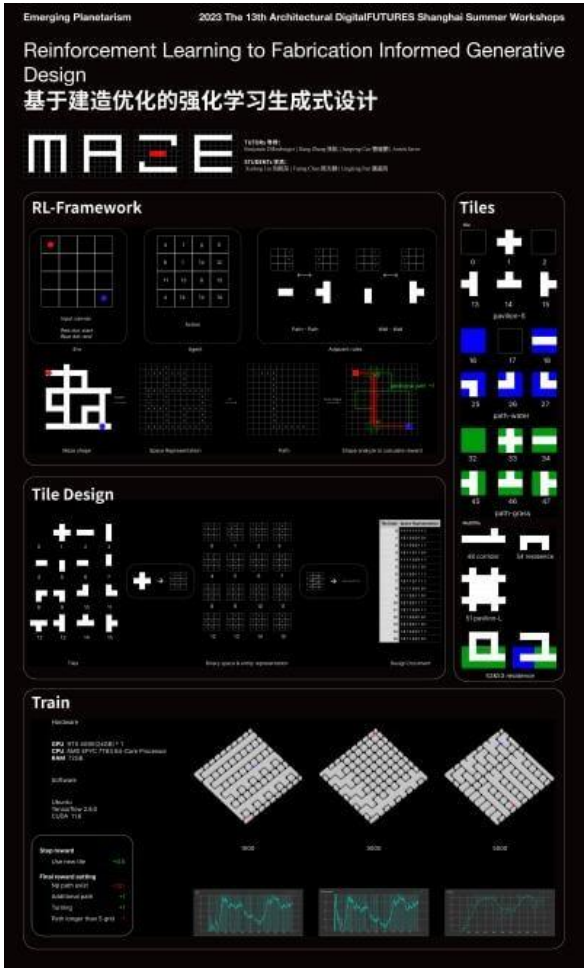
NONLINEAR SPACE

Exploring curved assemblies



LANDSCAPE

Generating field-scale organizations



ADJACENCY

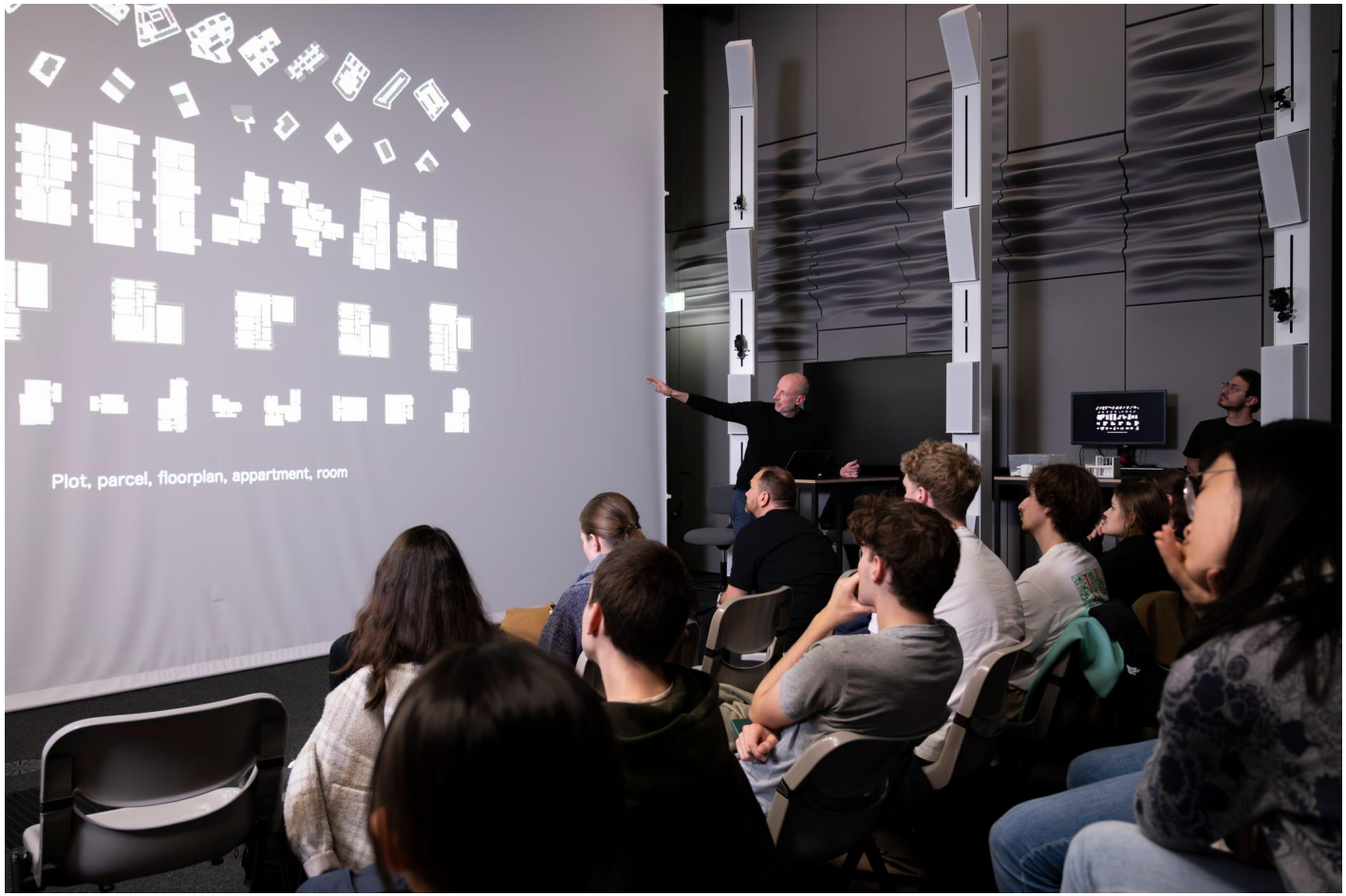
Encoding spatial relations as rules

TOP-UP · HS24 Seminar Week

Rooftop living through computational design and timber construction

21–25 October 2024 · ETH Zurich

Fourteen students explored modular timber rooftop extensions through computational design, a kit of parts, physical models, and iterative critique.



Briefing on computational housing precedents and rooftop design.

TOP-UP · Workshop Framing

From housing systems to a shared design brief

TOP-UP Seminar Week · ETH Zurich



Course framing connects configurable homes to industrialized production.

TOP-UP · Week Schedule

Five days of tools, kit design, prototyping, and review

Seminar plan · 21–25 October 2024

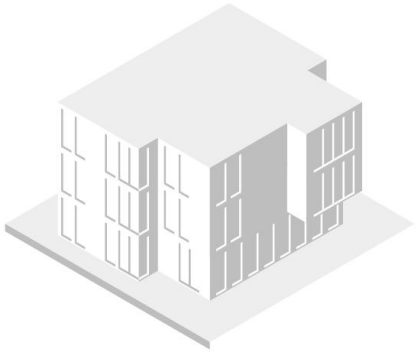
START	DURATION	ITEM	MONDAY (21.10)	TUESDAY (22.10)	WEDNESDAY (23.10)	THURSDAY (24.10)	FRIDAY (25.10)
8:45	1:45	block 1 (HIT F12)	**intro, brief, team formation (Anton Savov)**	Project idea, references, sketches	Generate with custom Kit-of-Parts	Adjust KoP, prepare deliverables	3D prints and finalize works
10:30	0:15	break	break	break	break	break	break
10:45	1:45	block 2 (HIT F12)	tools setup	Modify Kit-of-Parts	Generate with custom Kit-of-Parts	Adjust KoP, prepare deliverables	3D prints and finalize works
12:30	1:15	lunch	lunch	lunch	lunch	lunch	lunch
13:45	0:45	block 3 (HIT F12)	Generate and play with example	**input talk: Modularization (Dr. Jianpeng Cao)**	**Deskcrit w. ERNE, 2pm-4:30pm**	3D prints and finalize works	presentation setup
14:30	0:15	break		break			move to IDL
14:45	0:45	block 4 (HIT F12)		**deskcrit with Prof. Dillenburger and Prof. Hall**			**Student Presentations (5x a 20min) in IDL: Dillenburger, Hall, ERNE**
15:30	0:30	break	move to IDL			break	
16:00	**0:45**	**input talk (HIT F22)**	**From House to Product (Prof. B. Dillenburger)**				
16:45	**0:45**	**input talk (HIT F22 IDL)**	**Industrialized Constr. & Product Platforms (Prof. Daniel Hall)**		**input talk ERNE: Digital Timber Prefab**	internal pinup	
17:30	0:30	Outlook	Daily Insights Round	Daily Insights Round	Daily Insights Round		**Apero (till 19:00)**
18:00	end						

The seminar moves from team formation and tool setup to custom kits, desk critiques, models, and final presentations.

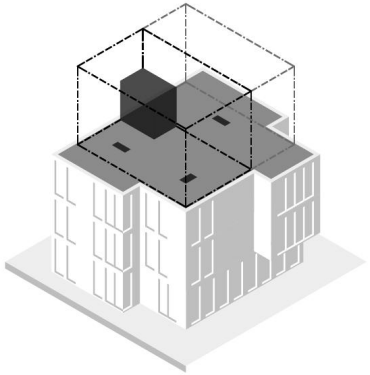
TOP-UP · Design Workflow

A common sequence for site-specific rooftop proposals

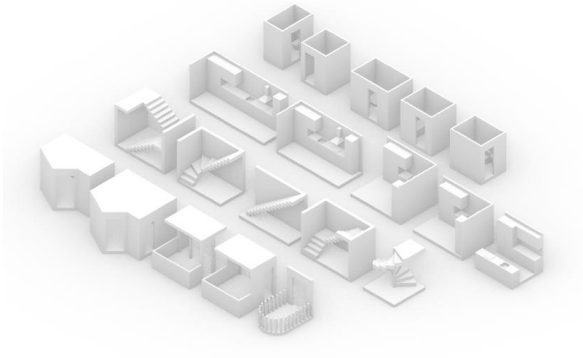
Workshop method



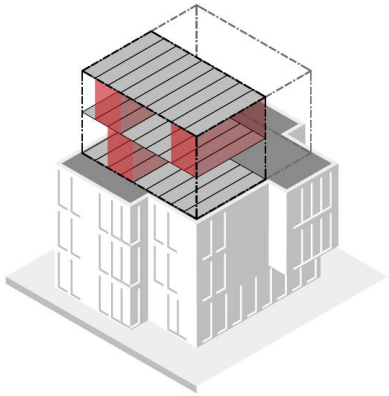
01 Pick a building



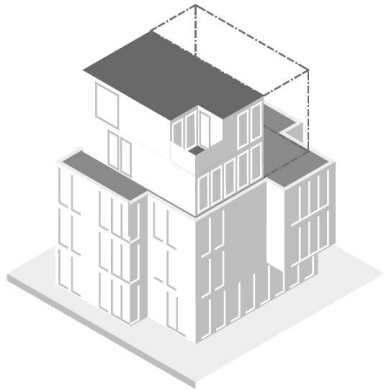
02 Assess the roof extension



03 Define a kit of parts



04 Generate and review layouts



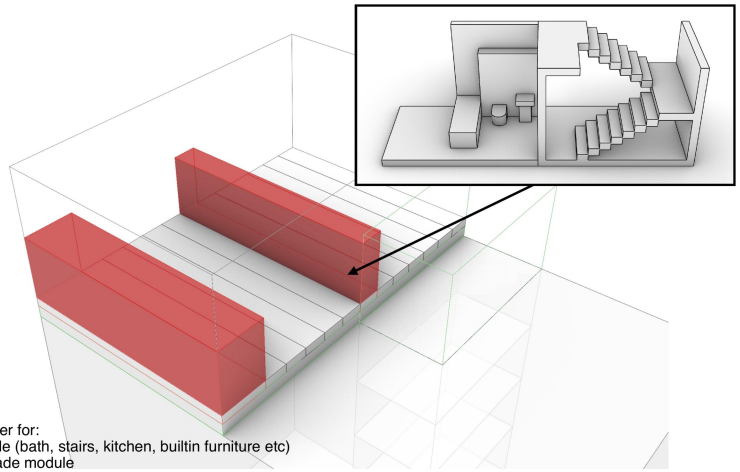
05 Preview the design

TOP-UP · Assembly Logic

Service modules, wall products, and a second floor

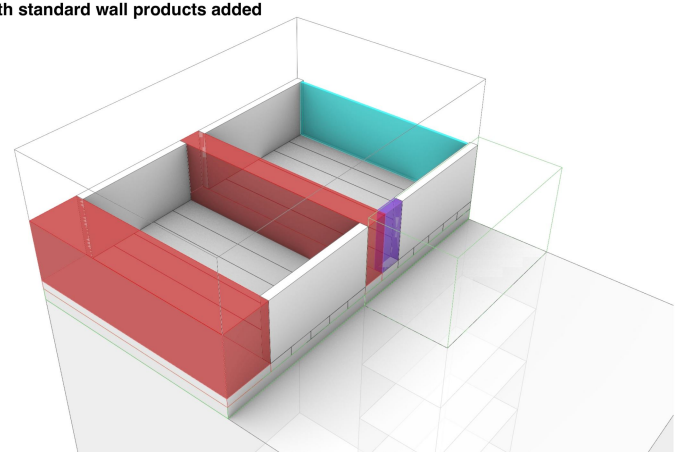
Workshop design example

Assembly

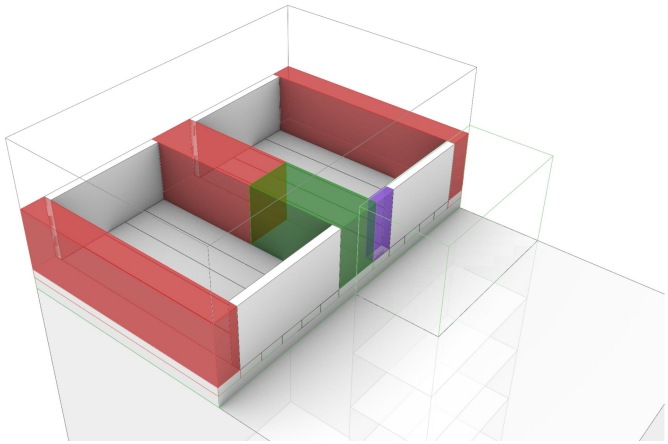


Red is a placeholder for:
- services module (bath, stairs, kitchen, builtin furniture etc)
- volumetric facade module

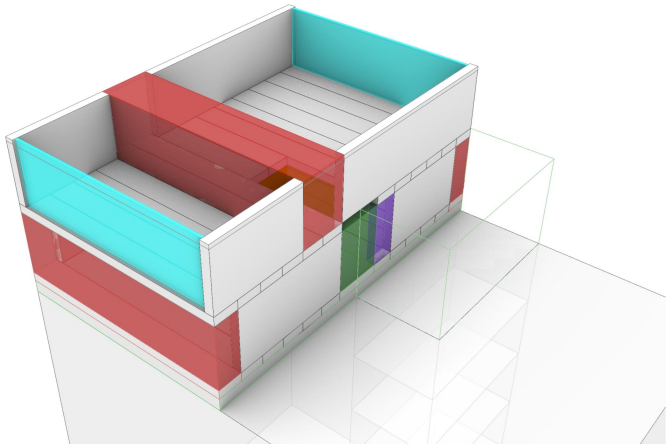
Assemblies with standard wall products added



Other option



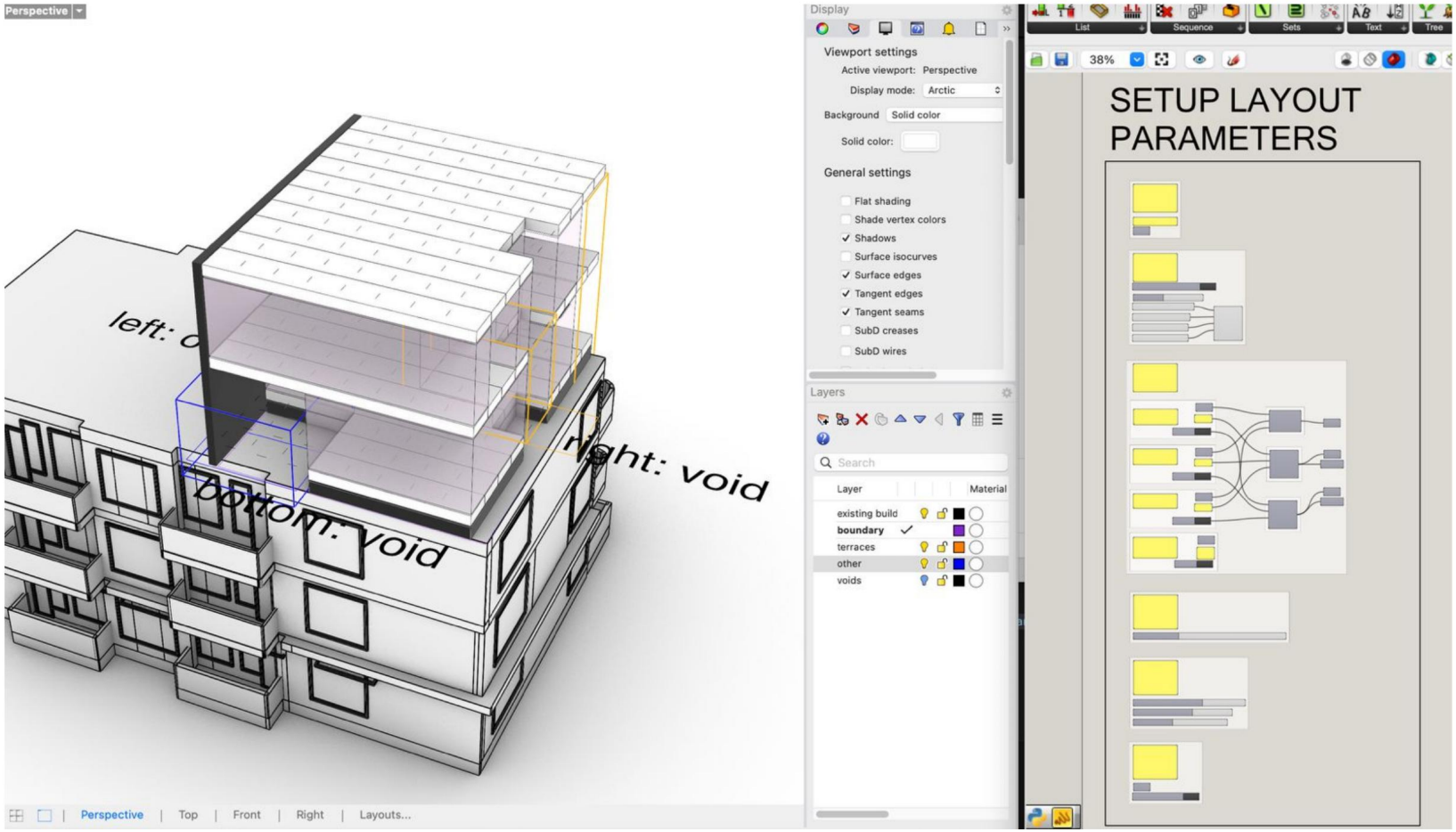
Second floor



TOP-UP · Setup Layout

Parameterizing the roof and extension boundary

Workshop toolchain

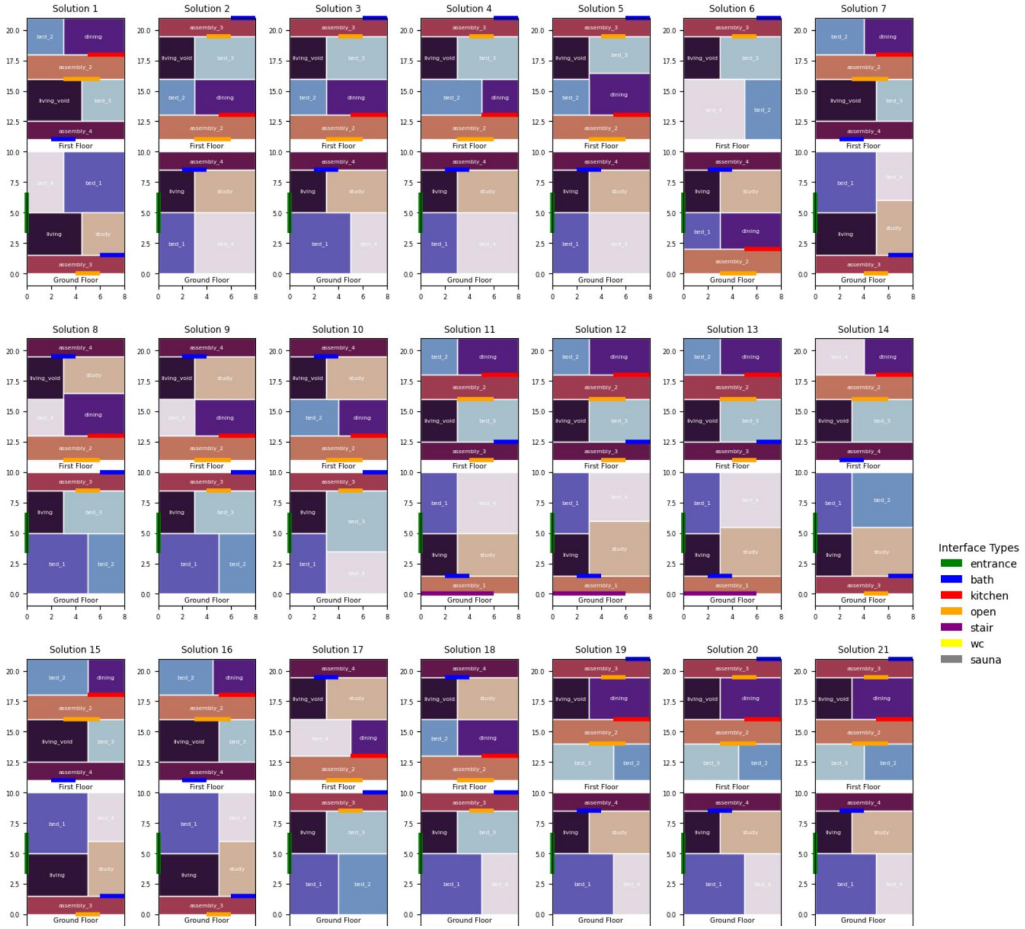


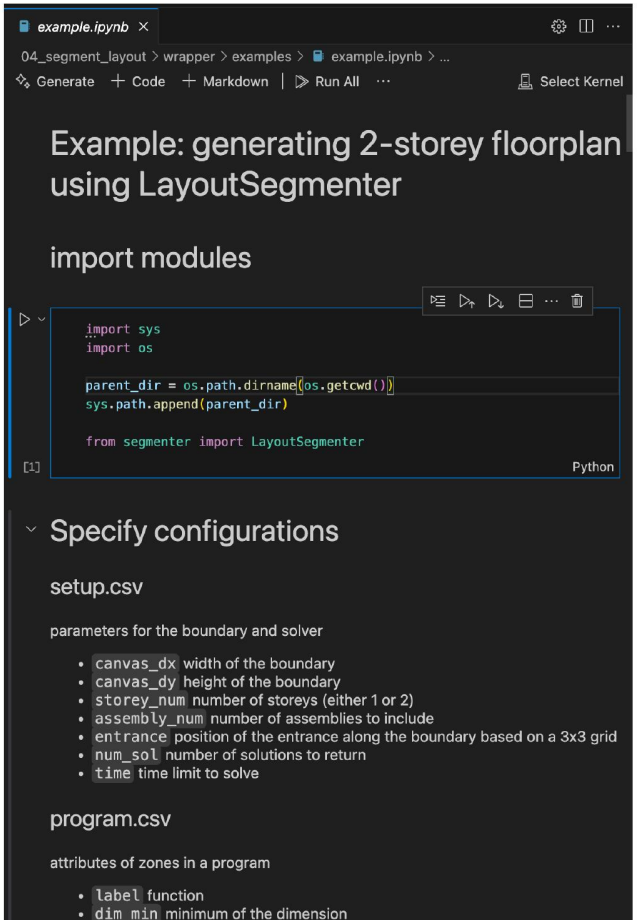
A Rhino model and Grasshopper controls define the available roof volume and layout conditions.

TOP-UP · Generate Layouts

Reviewing alternative two-storey arrangements

Workshop toolchain



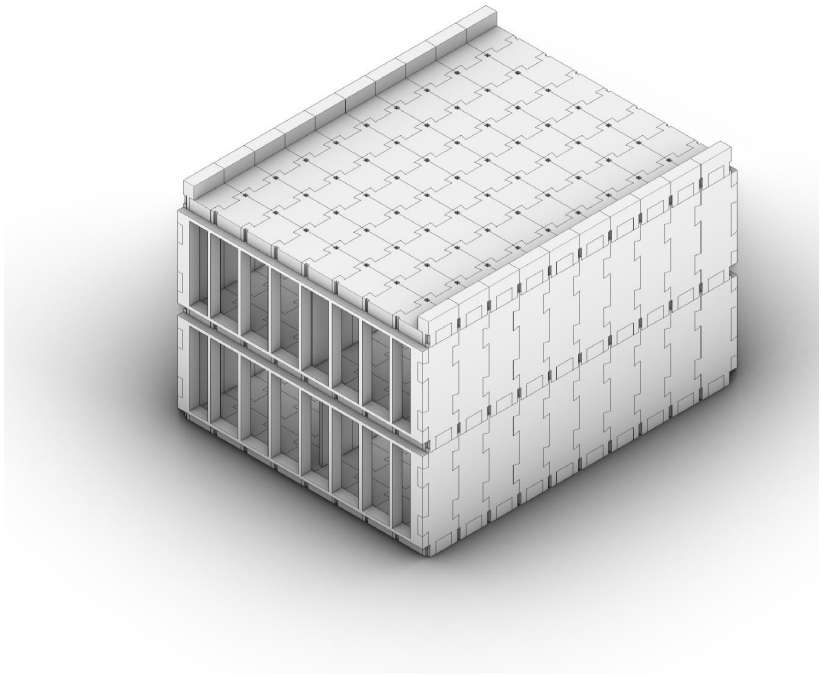


The LayoutSegmenter example explores layout options from boundary, program, and module inputs.

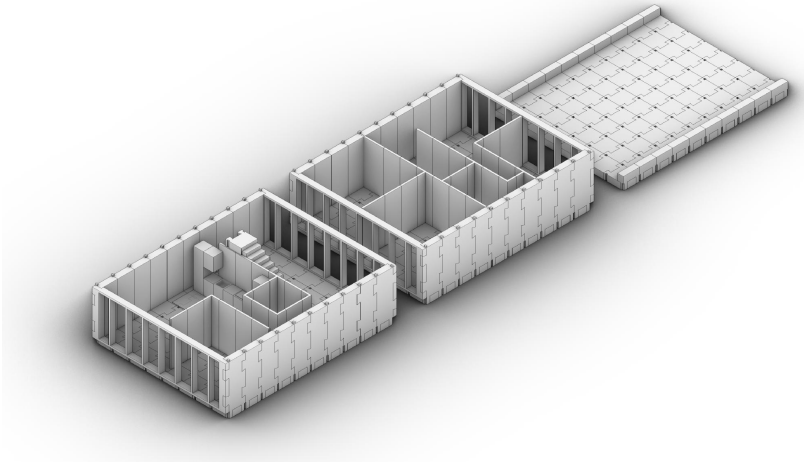
TOP-UP · Kit of Parts

An assembly, its spatial interior, and a component inventory

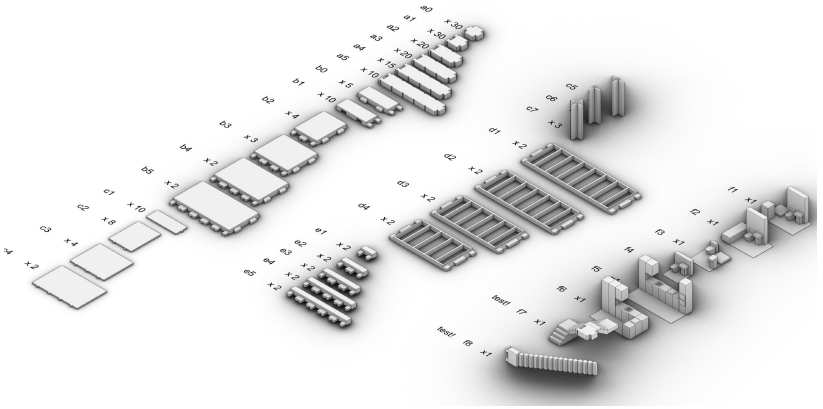
Workshop design example



Enclosed unit



Interior arrangement



Component inventory

TOP-UP · Collaboration

Students test shared design variables and tool outputs

Workshop documentation



Team briefing around a computational design model.



Reviewing the generated options in a small group.

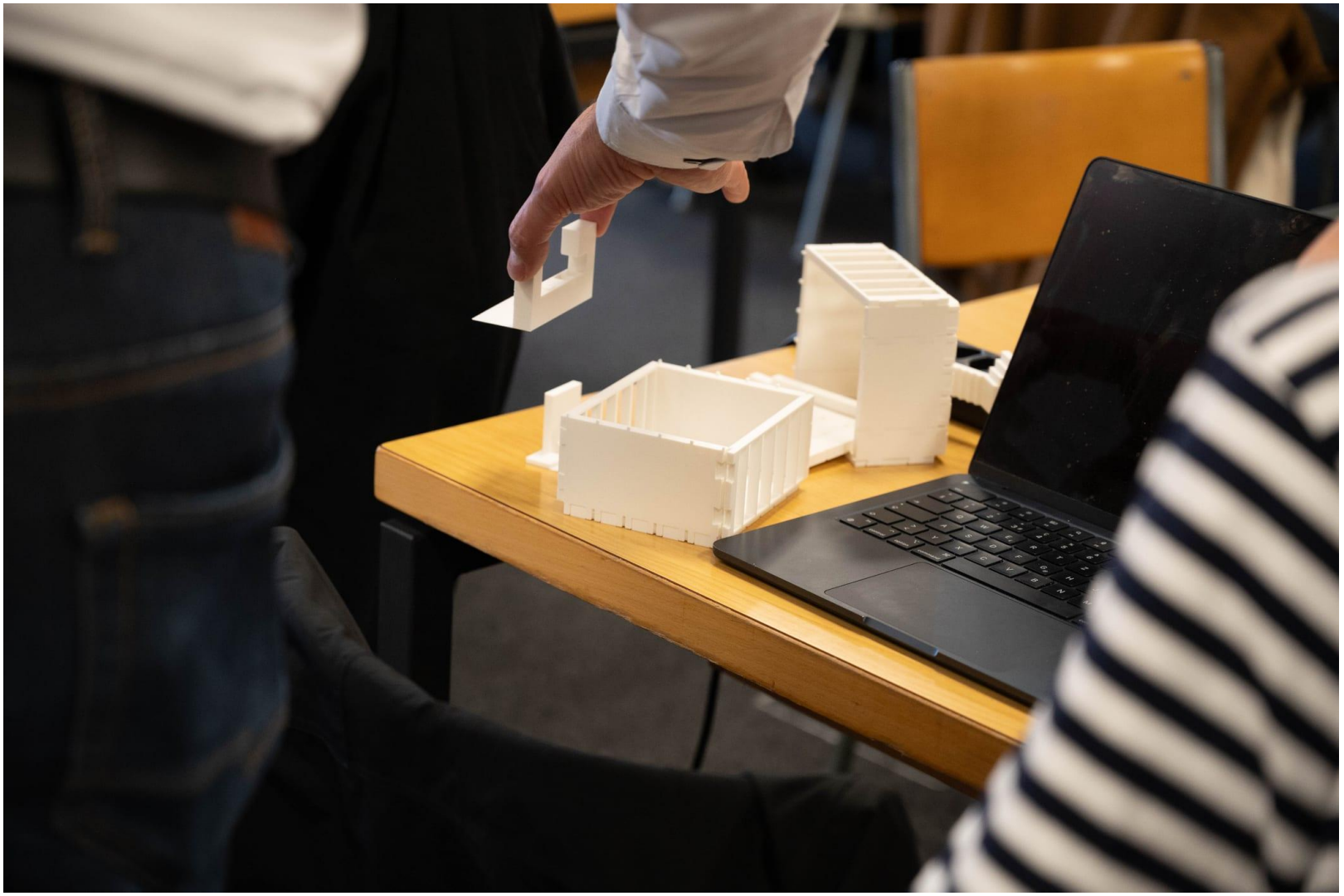


Digital layouts are checked against physical prototypes.

TOP-UP · Prototyping

Physical models expose assembly choices

Workshop documentation



Students compare spatial options at model scale.

TOP-UP · Final Review

Teams explain how layouts meet construction logic

Workshop documentation



A team presents its rooftop proposal.



Students present the layout logic.

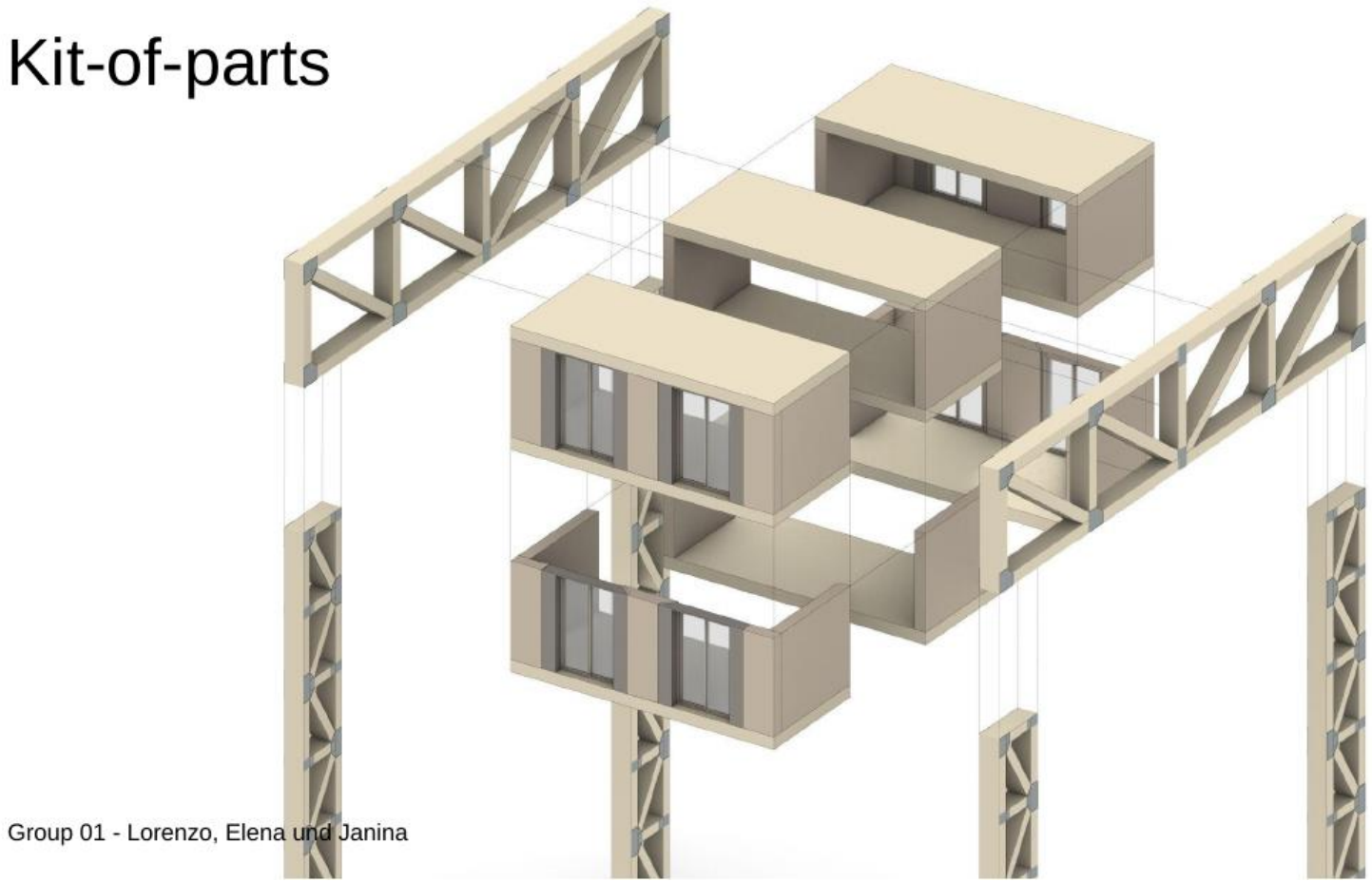


Reviewers question the prefabrication strategy.

Student Work · Truxes

Timber trusses carry adaptable rooftop boxes

Group 01 · Lorenzo, Elena, and Janina



The kit separates a supporting truss frame from modular living volumes.

Truxes · Rooftop Proposal

A modular extension above the existing building

Group 01 · Lorenzo, Elena, and Janina



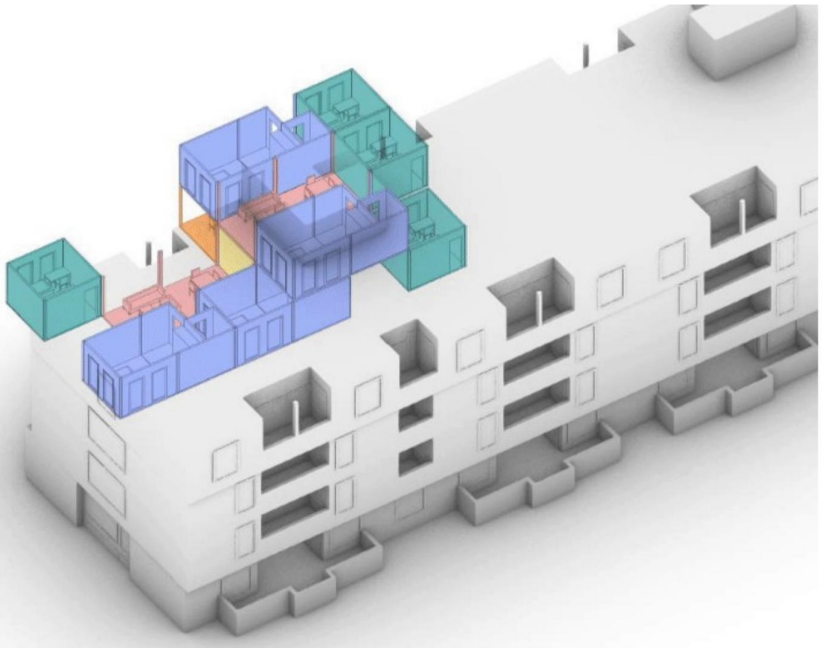
The proposal places timber-framed housing on the roof while leaving the existing structure legible.

Student Work · Living in Boxes

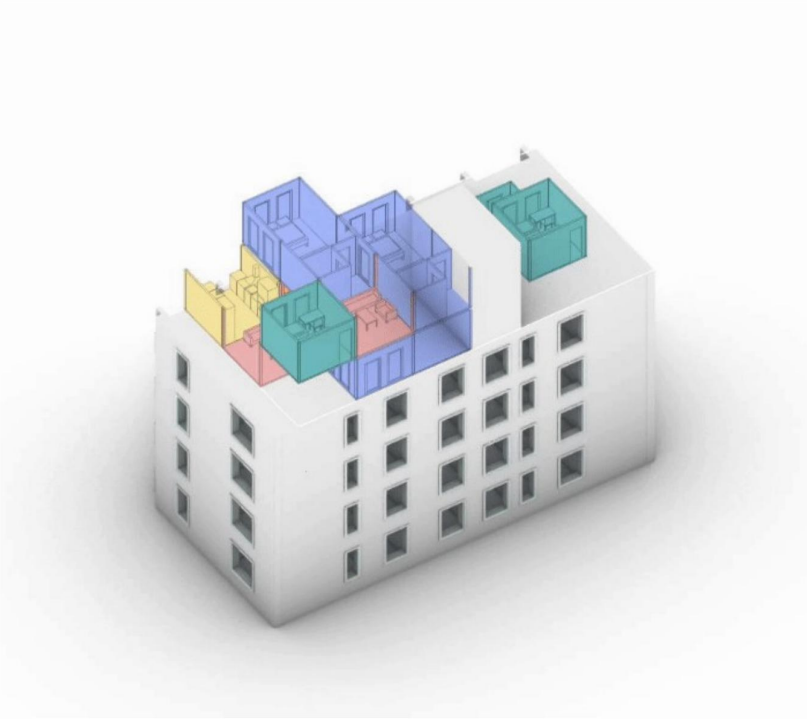
Customized modules organize rooftop clusters

Group 02 · Quentin Wiesmath, Congyi Ling, Jaime Ramos, Laura Ciak

modules building 1



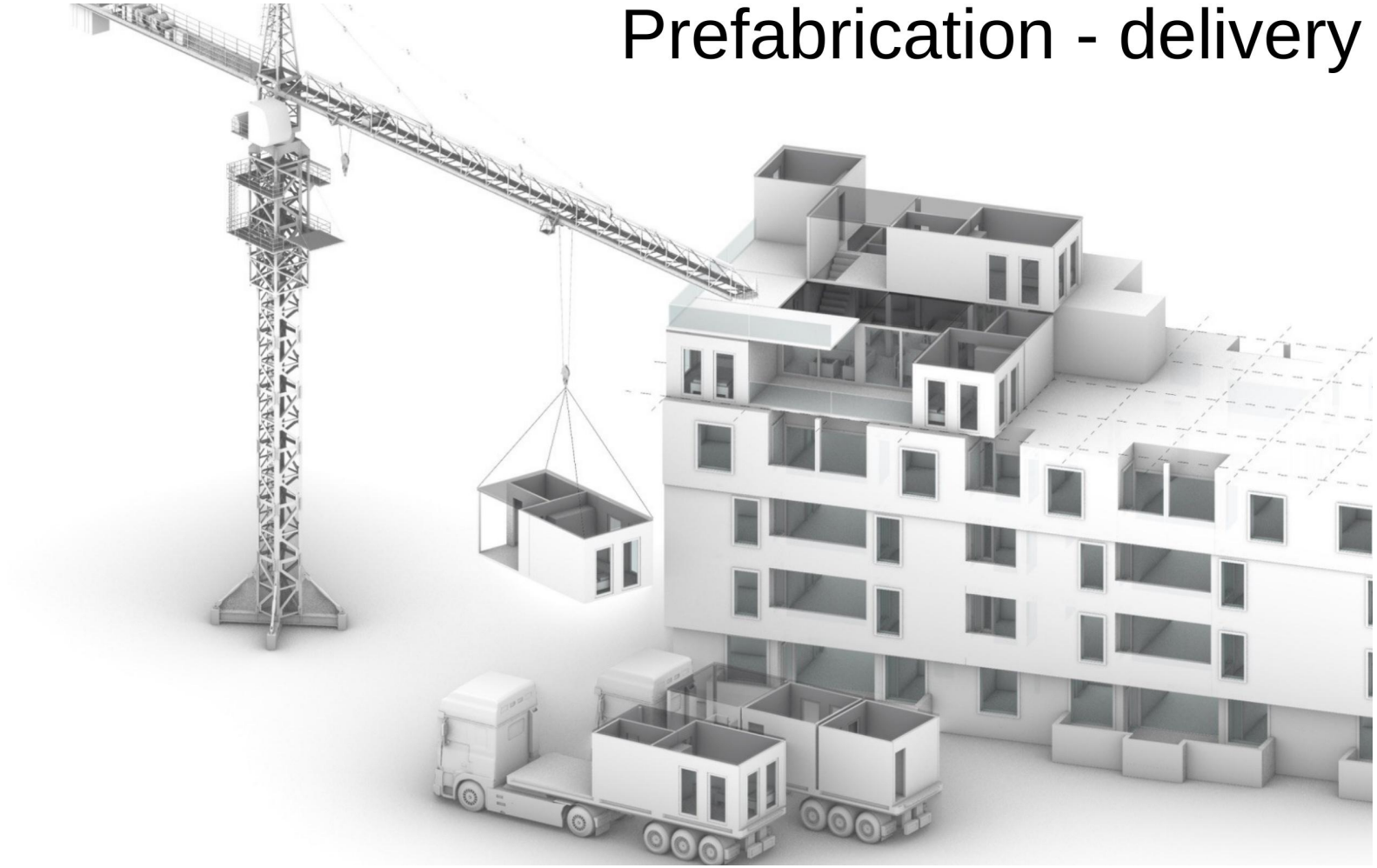
modules building 2



Living in Boxes · Delivery

Prefabricated units are transported and lifted into place

Group 02 · Quentin Wiesmath, Congyi Ling, Jaime Ramos, Laura Ciak

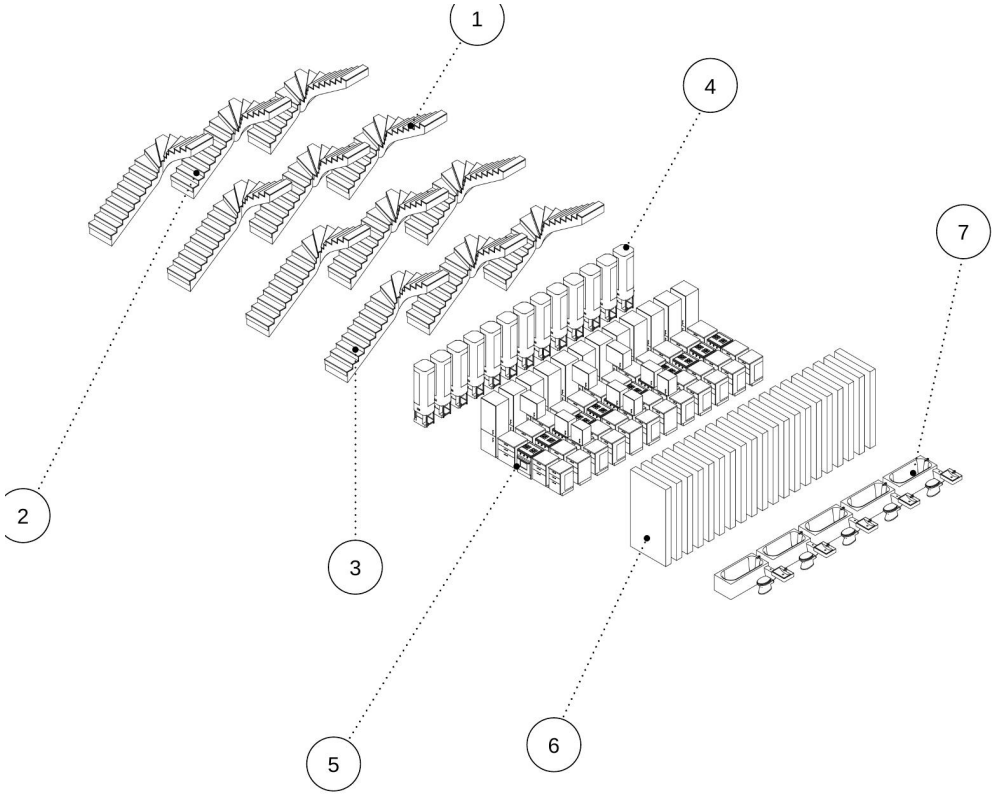


The proposal connects modular design to production, transport, and on-site assembly.

Student Work · Modular Nucleus

Factory-prepared stairs, services, and wall elements

Anja Meyner, Michelle Bingisser, Gianna Pirovino, Théo Droux



Stock of Elements in Factory, ready to be collected and assembled

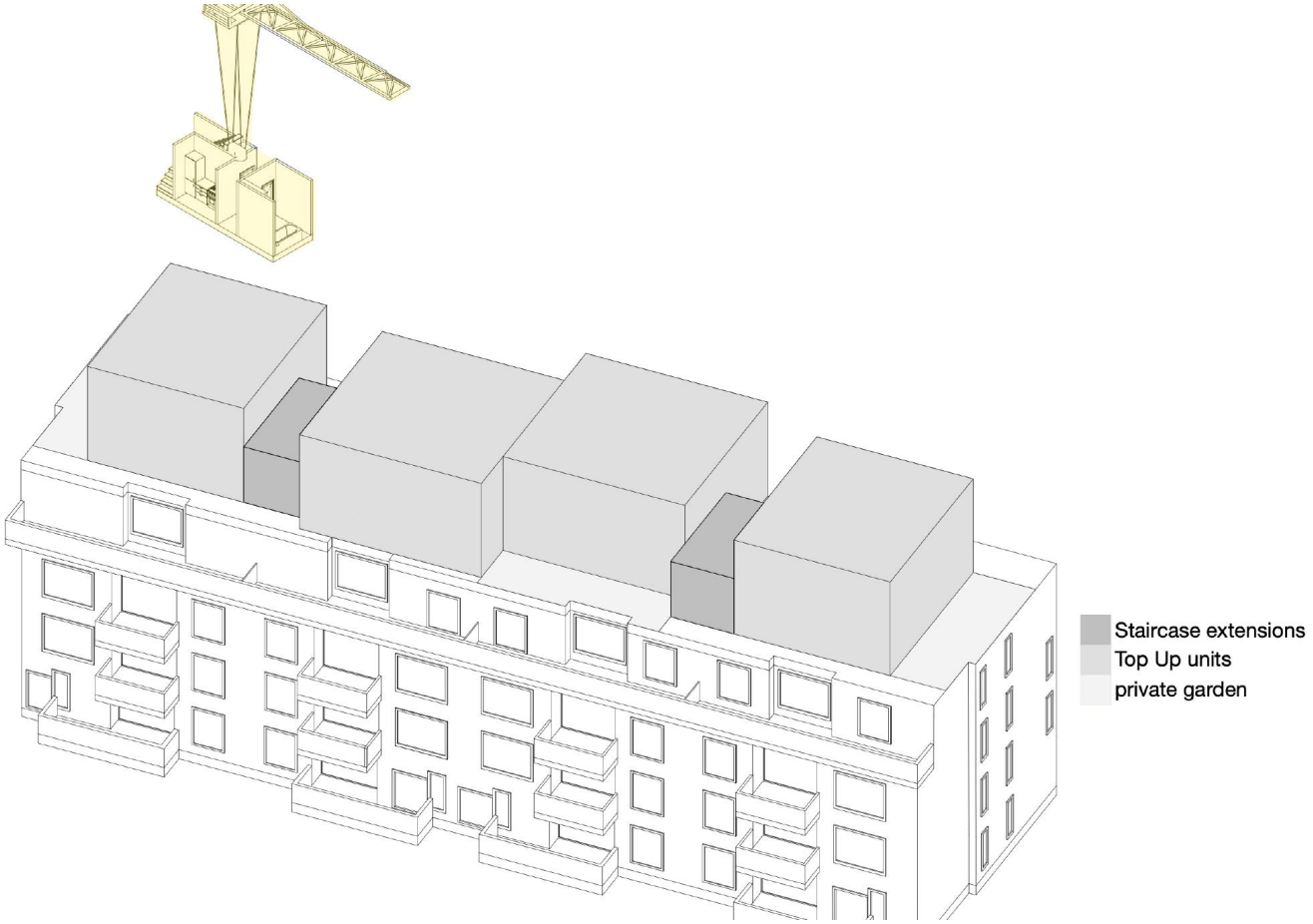
1. Stair Option 1
1. Stair Option 2
1. Stair Option 3
1. Heat Pump
1. Kitchen Parts
1. Vorwand
1. Bathroom Elements

A stock of functional parts supports configurable rooftop units.

Modular Nucleus · Roof Installation

Assembled units are lifted onto the existing building

Anja Meyner, Michelle Bingisser, Gianna Pirovino, Théo Droux

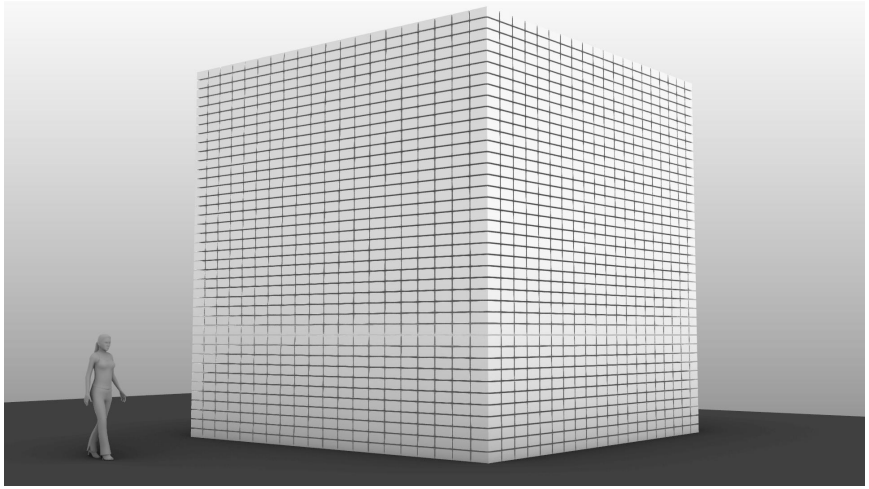


The final axonometric links factory assembly to rooftop installation.

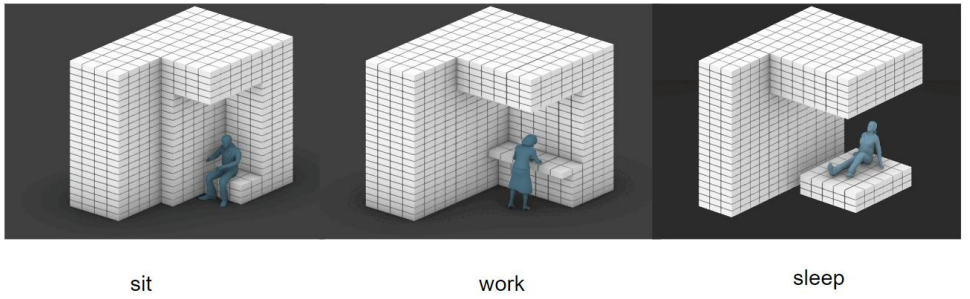
MAS DFAB · Rewarded Affordance

Voxelized space and action prototypes for deep reinforcement learning

Student: Yixiao Huang · Tutor: Hang Zhang · ETH Zurich, 2023



Initial voxel field



Exploration, navigation, and everyday affordance actions

The agent carves a voxel field with actions tied to movement and daily use. Spatial affordance supplies the design objective.

Rewarded Affordance · Reward Design

Training balances space carving with use and movement

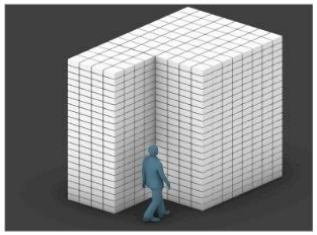
Yixiao Huang · MAS DFAB thesis

Reward

At Termination

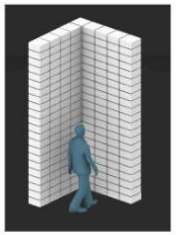
Complete affordances
Reward **+10/ affordance**
Penalty **-10/ exceeded**

E.g. 5 seats are desired, 4 are created, +10*4 is received when terminate



Keep turning consecutively
Penalty **-20**

Keep turning is considered inefficient and will slow down training

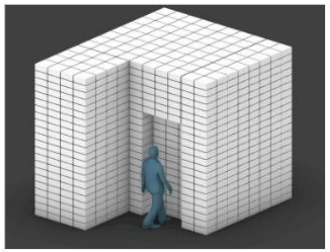


At every step

Carve out space
Reward **+0.02/ voxel**

Exploring the space is encouraged

E.g. taking walk action carves out 48 voxels, thus 0.02*48 = 0.96 reward given



Cost per step
Penalty **-0.3/step**

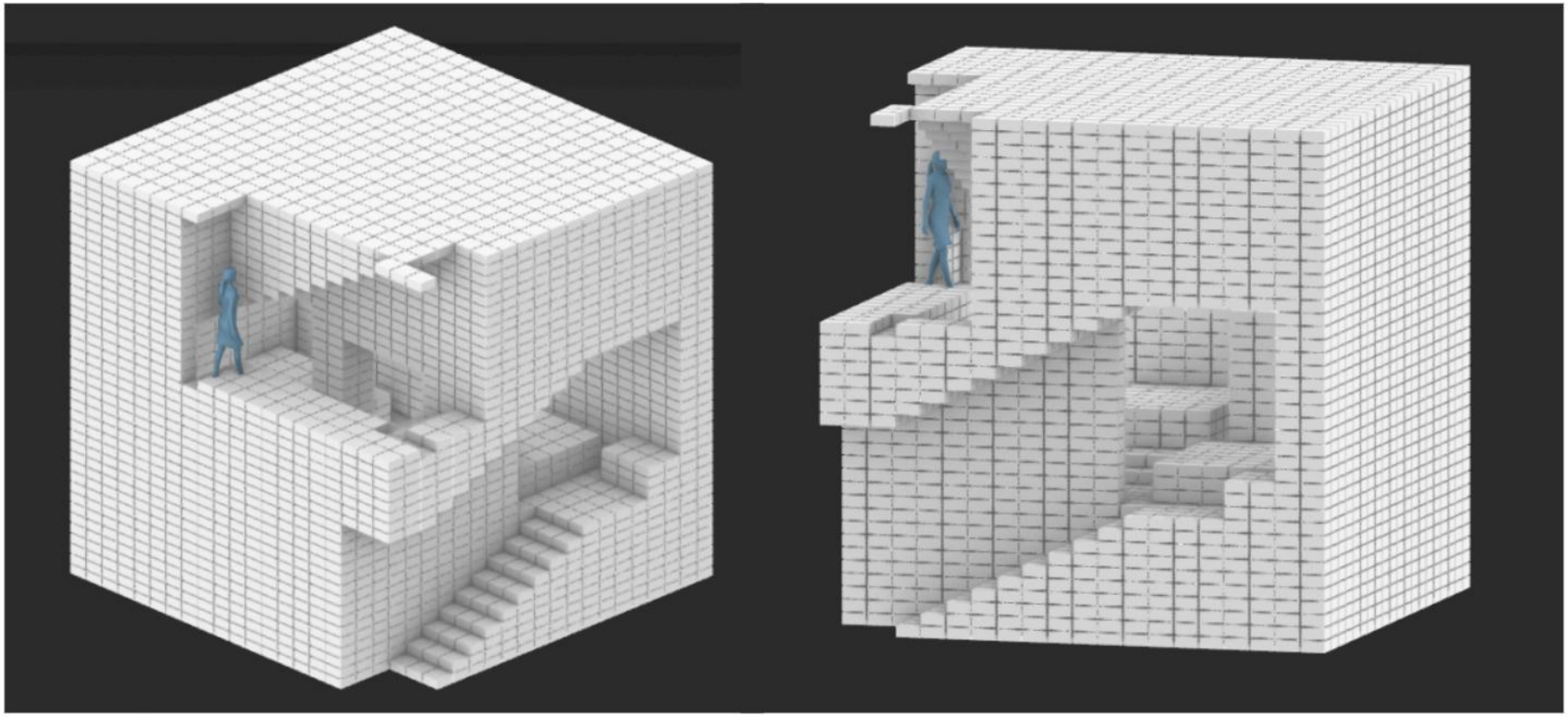
The agent is exploring the space with a cost

The reward setup values completed affordances and exploratory carving while penalizing repetitive actions.

Rewarded Affordance · Spatial Results

A composite model generates carved living spaces

Yixiao Huang · MAS DFAB thesis

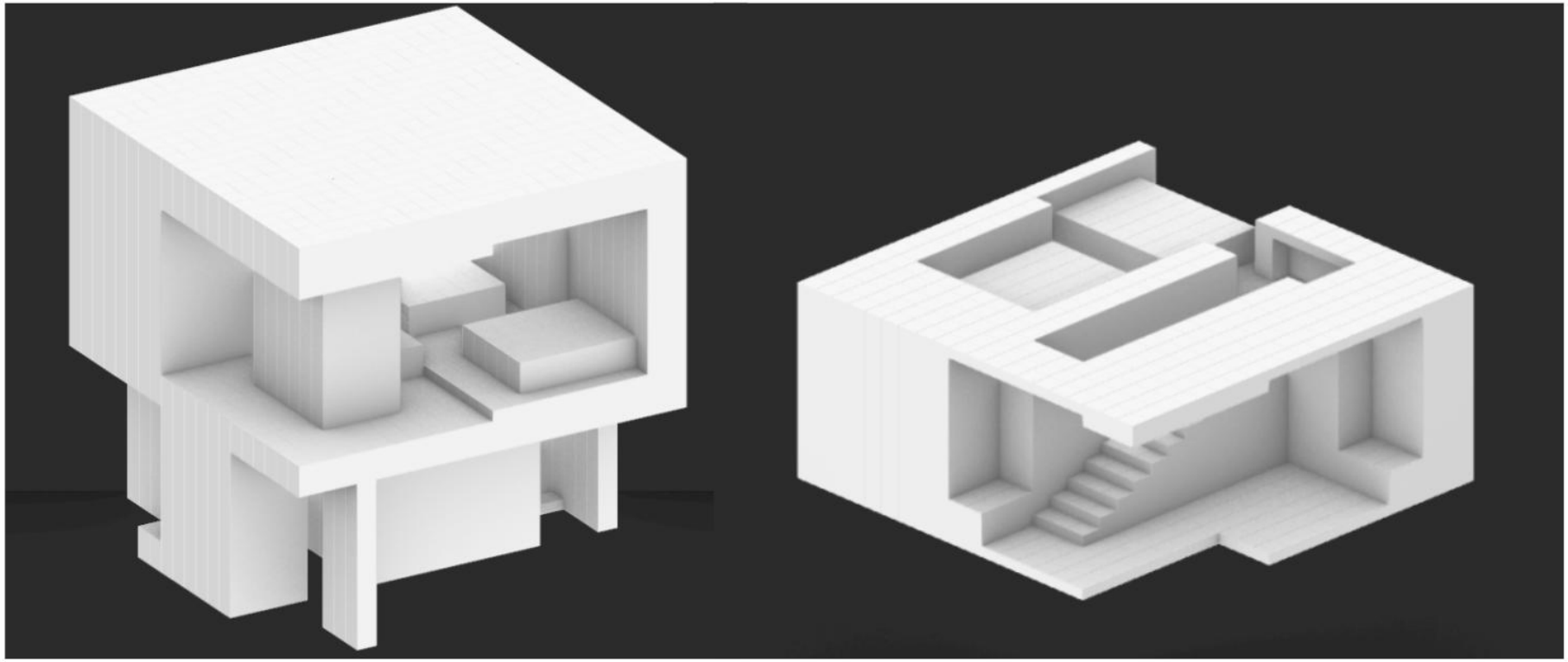


A two-storey spatial configuration combines voids, circulation, and rooms.

Rewarded Affordance · Variations

Later experiments explore different connected spaces

Yixiao Huang · MAS DFAB thesis

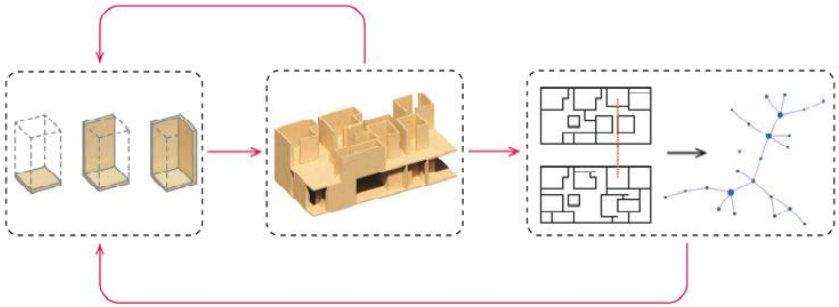


The source compares connected and separated outcomes from a multi-agent setup.

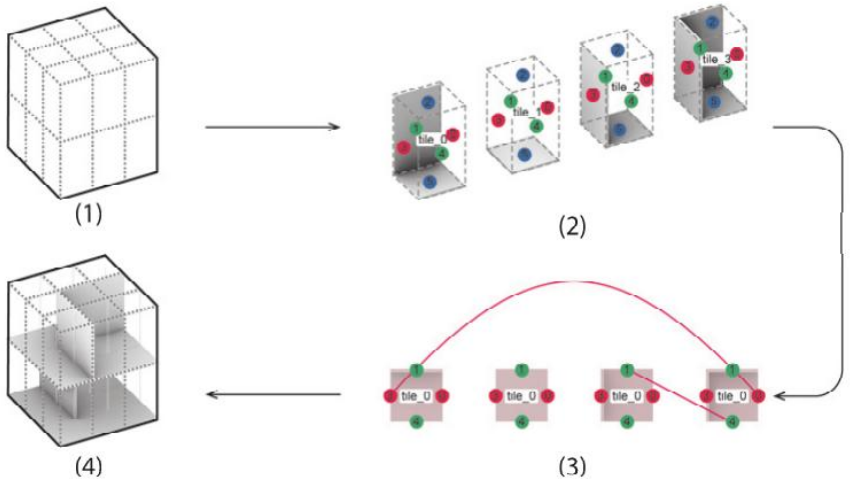
MAS DFAB · Architectural Tilesets

Generation and evaluation guide the tile design

Hanbing Zhao · with Anton Savov, Hang Zhang, Benjamin Dillenburger · eCAADe 41 (2023)



Tile families are assembled, evaluated, and revised.

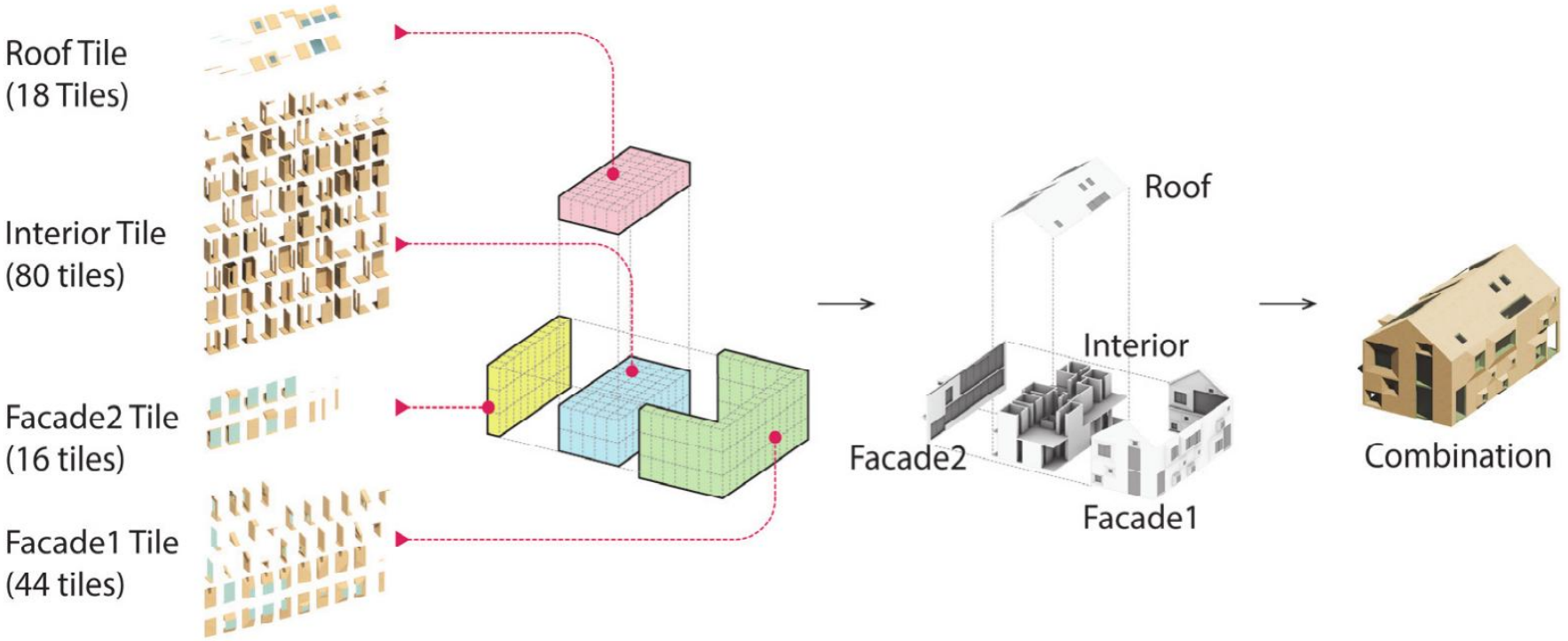


Wave Function Collapse assigns tiles while matching edge constraints.

Architectural Tilesets · Generation

Roof, interior, and facade tiles combine into a house

Hanbing Zhao · with Anton Savov, Hang Zhang, Benjamin Dillenburger · eCAADe 41 (2023)

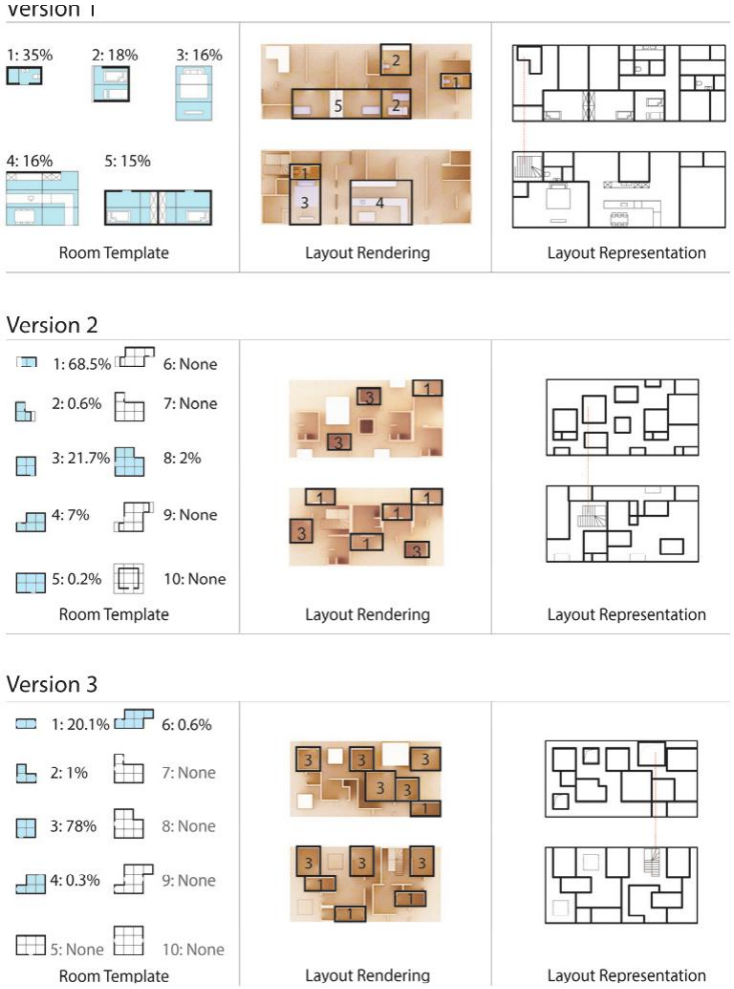


Roof, interior, and facade tiles enter a Wave Function Collapse workflow to produce house combinations.

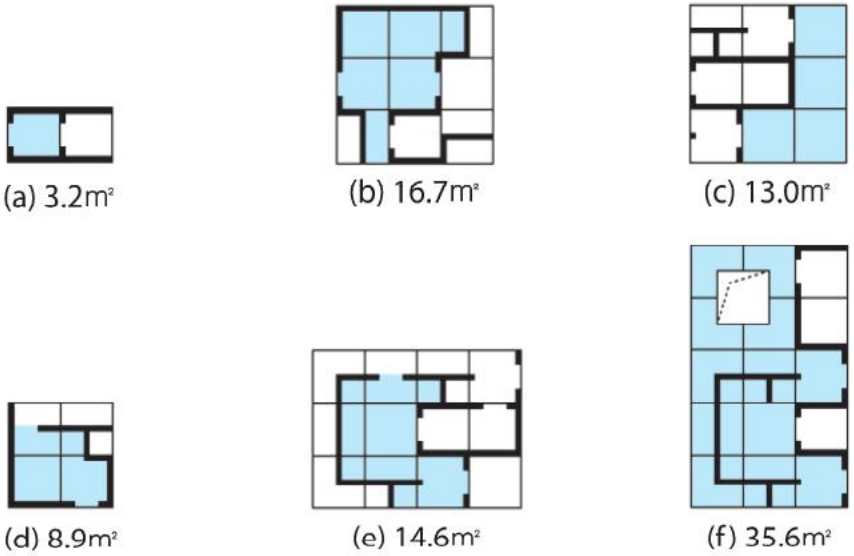
Architectural Tilesets · Design Iteration

Three tileset versions change room templates and layouts

Hanbing Zhao · MAS DFAB series



Three tileset versions



Examples of room shapes from the third version

The design process adjusts tile forms and rules, then checks the range and quality of generated rooms.

Undergraduate Teaching

Computational Design II–IV at ETH Zurich

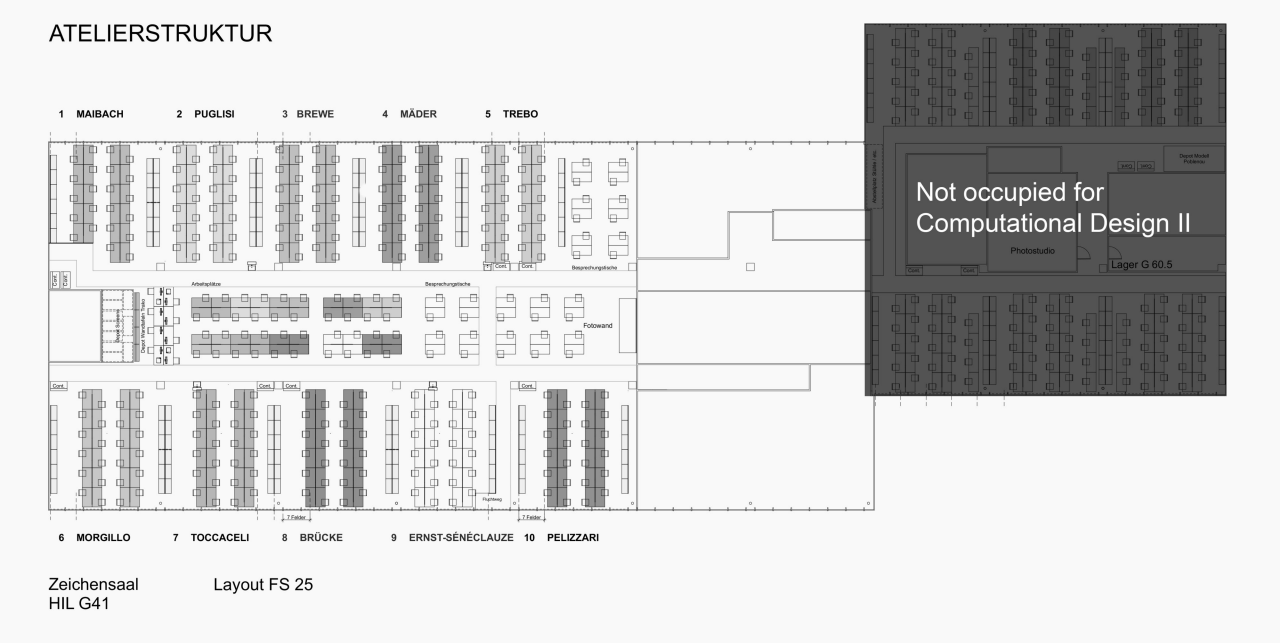
Teaching team member · FS2025, HS2025 and FS2026

MY TEACHING CONTRIBUTION

Teaching selected tutorials and exercise sessions.

Grading assignments.

Preparing selected course materials and exam content.



FS2025 studio layout for ten tutor-supported flipped-classroom ateliers.

TEACHING FORMAT

Lectures establish concepts. Tutorials and supported exercises translate them into design workflows. Assignments connect technical decisions to architectural intent and reflection.

COMPUTATIONAL DESIGN II

FS2025
Code, analysis, search and digital fabrication.

COMPUTATIONAL DESIGN III

HS2025
Generative design, AI-assisted coding and visualization.

COMPUTATIONAL DESIGN IV

FS2026
Point clouds, existing buildings and physical prototyping.

Computational Design II · Course Structure

Four linked blocks introduce computational methods through design assignments

FS2025 · 052-0606-00L · ETH Zurich

Each block combines a lecture, tutorials and an assignment. Flipped-classroom sessions provide tutor assistance. Moodle hosts the learning materials, questions and submissions.

Block	Computational method	Student task	Submission evidence
01 Code & Procedural Design	Python variables, lists and loops in Grasshopper	Generate 2D patterns through iterative geometry operations	Line-drawing poster, explanation, script and workflow files
02 The Data of Form	Mesh normals, ray casting and geometric analysis	Evaluate a design and visualize the findings	False-colour poster, analytical reflection and workflow files
03 Search & Optimize	Parametric massing and evolutionary search with Galapagos	Translate design goals into measurable fitness criteria	Selected alternatives, rationale and workflow files
04 Machine Translations	Layered patterns and laser cutting with RapLab	Design and fabricate an A5 surface with at least three layers	Physical model photograph, cut file and design reflection

Software and teaching materials include Rhino, Grasshopper, Python and Blender. Each assignment asks students to connect a technique to a design intention.

Code & Procedural Design

Python loops extend the generative logic of Grasshopper

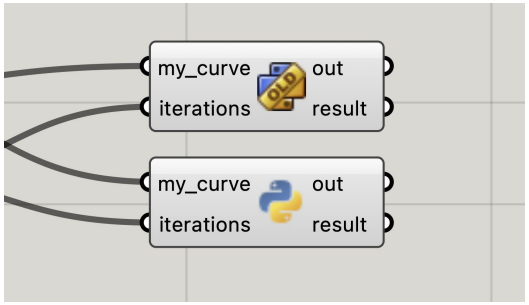
Computational Design II · Assignment 1 · FS2025

ASSIGNMENT

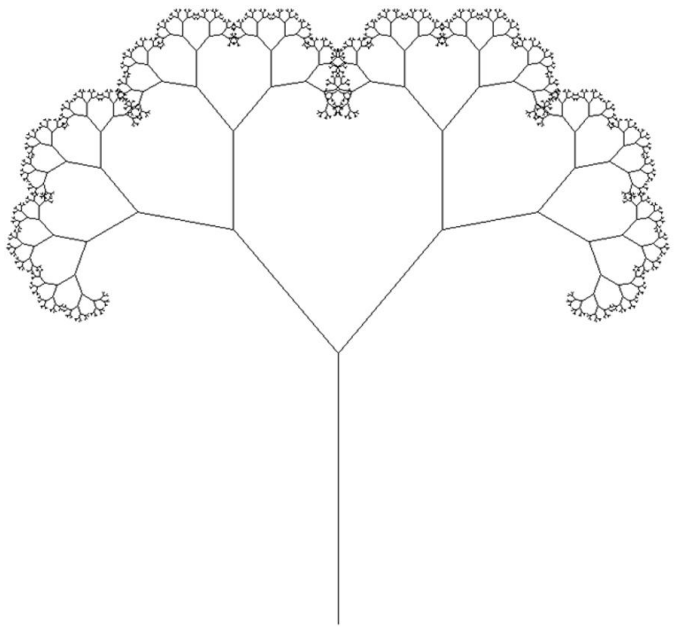
Generate a 2D pattern by reusing or transforming geometry from the preceding loop iteration. Explore a theme through controlled experiments with input variables.

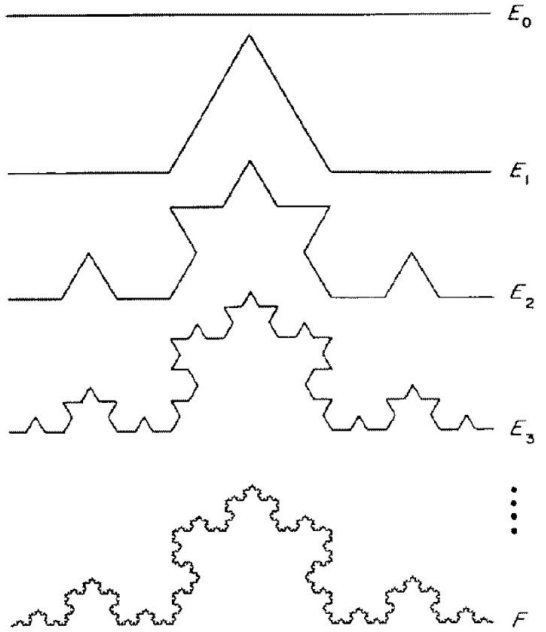
TECHNICAL WORKFLOW

Adapt the supplied Python component. Use variables, lists and a for loop. Control iterations with sliders or curves, compare versions, then bake the result into Rhino.

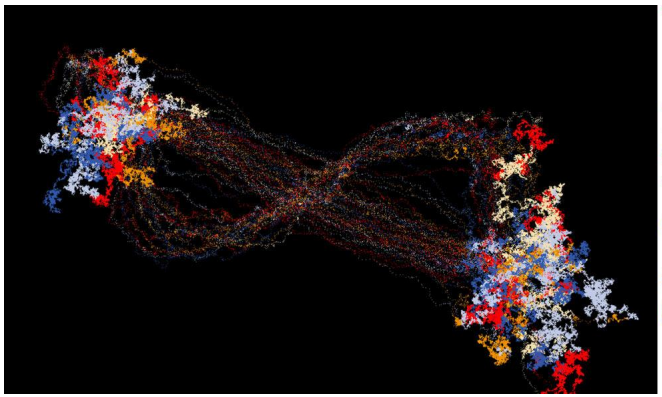


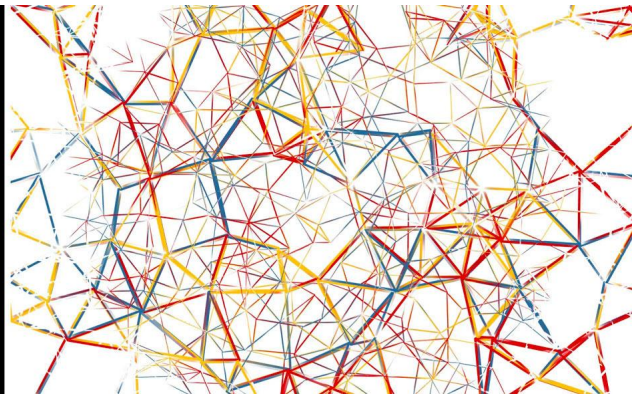
Rhino 8 Python 3 Script component shown alongside the earlier component.





Teaching references: Paul Bourke's L-system tree and the Koch snowflake illustrate recursive geometry.





Random-walk and Processing examples introduce visual experimentation through code.

The Data of Form

Geometric analysis helps students assess and revise architectural form

Computational Design II · Assignment 2 · FS2025

ASSIGNMENT

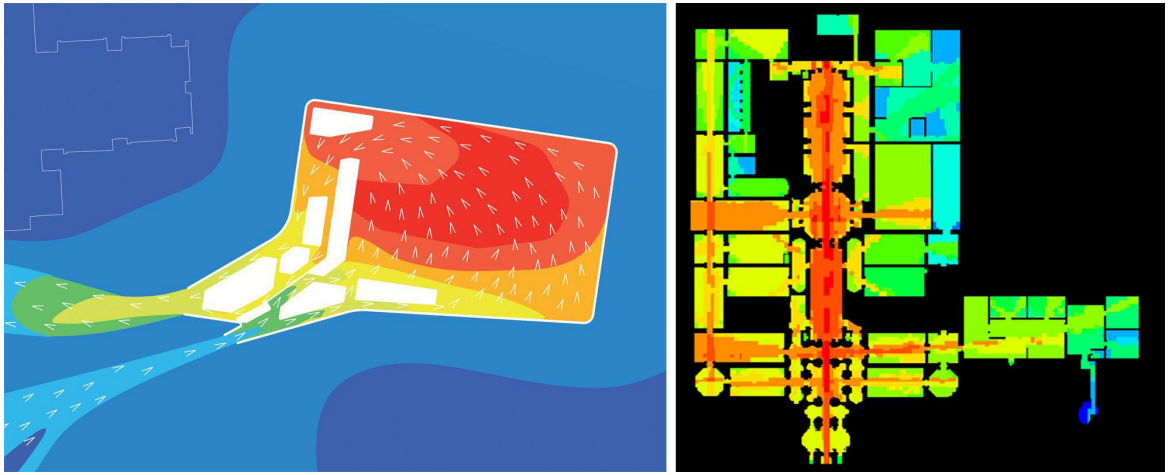
Analyze a mesh from an earlier design project. Ask how its geometry performs in relation to sunlight, orientation, facade area or another architectural concern.

TECHNICAL WORKFLOW

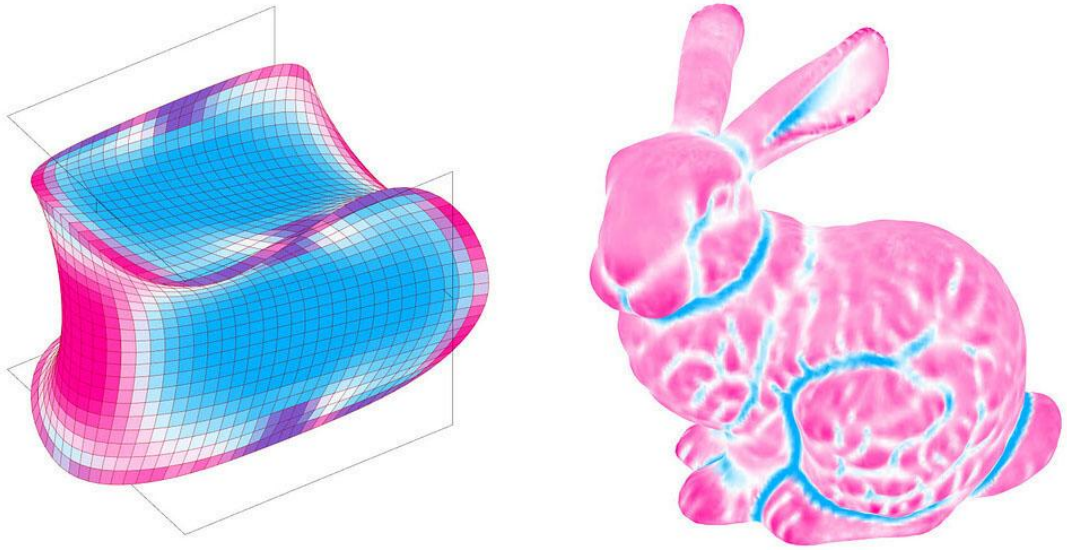
Prepare an evenly spaced mesh with quad remeshing. Apply normal-vector and ray-casting routines, bake a coloured mesh, and select a view that makes the finding clear.

REFLECTION

Explain why the analysis matters and whether it revealed a surprise, confirmed an intuition or suggested a design improvement.



Teaching references: Philippe Rahm architects and UCL Space Syntax Laboratory use analysis to investigate architectural performance.



Rhino and Grasshopper mesh-analysis examples use colour to reveal geometric properties.

Search & Optimize

Students encode architectural priorities in an evolutionary fitness function

Computational Design II · Assignment 3 · FS2025

ASSIGNMENT

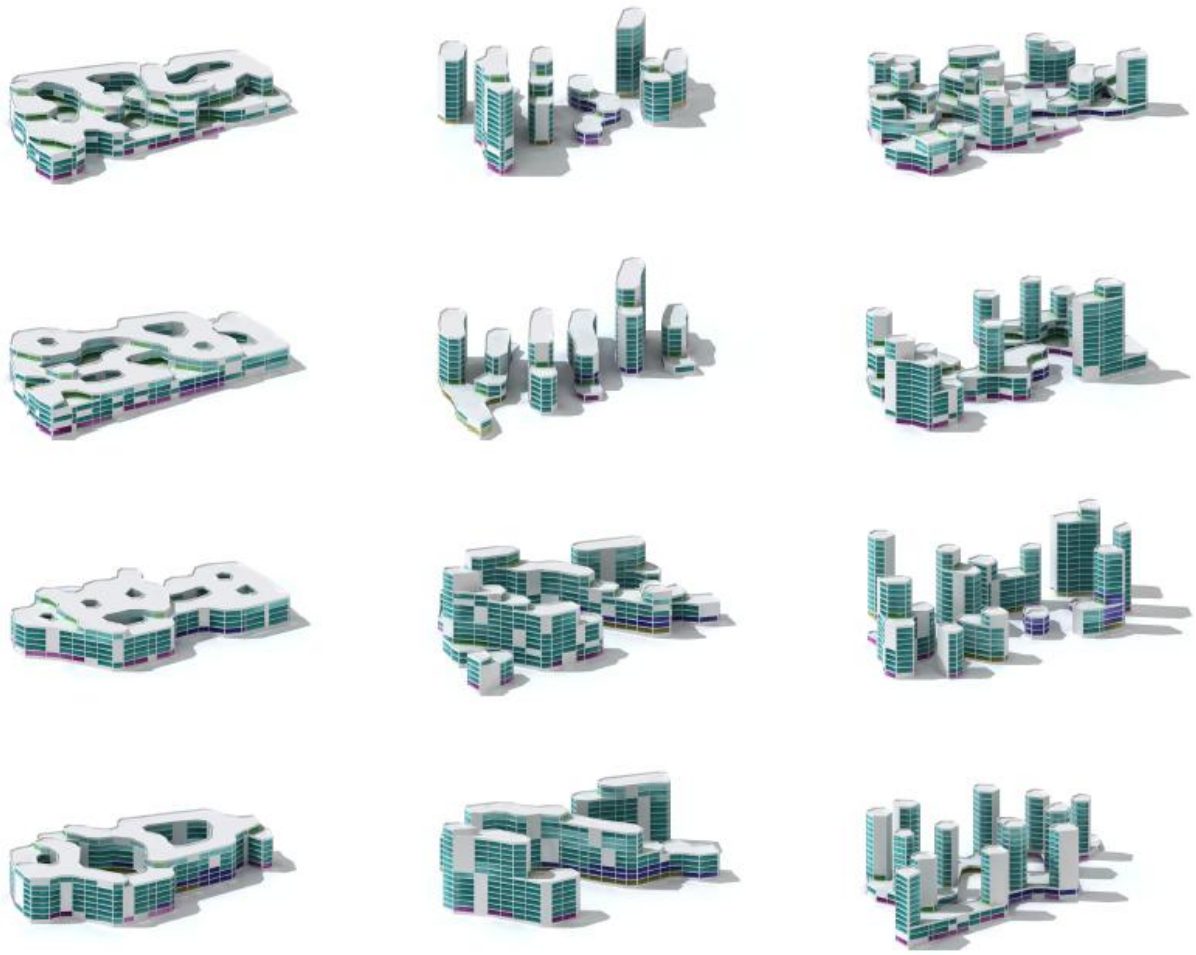
Build a parametric massing model with several volumes. Choose metrics such as facade surface, floor area or geometric complexity, and decide how to balance competing objectives.

ITERATIVE SEARCH

Run Galapagos repeatedly. Adjust the model, fitness criteria and genetic parameters. Save optimal and suboptimal alternatives to reveal the range of the design space.

DESIGN REASONING

Explain model variation, the choice of metrics and their effect on composition. Assessment considers geometric control, technique integration and understanding of fitness landscapes.



Teaching reference: Kaisersrot's Institutsgenerator illustrates alternatives between towers and courtyard typologies.

Machine Translations

Layered patterns become physical architectural surfaces through laser cutting

Computational Design II · Assignment 4 · FS2025 · RapLab collaboration

ASSIGNMENT

Design a surface with at least three material layers. Relate density, transparency and depth to a facade, ceiling or interior cladding concept.

DESIGN AND FABRICATION

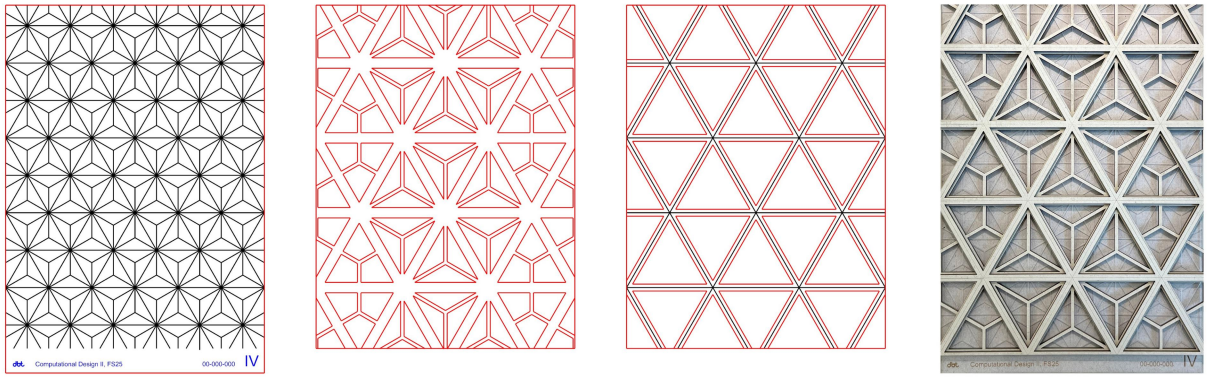
Create a base grid, filter geometry, then use offsets, subdivisions and infills. Plan the hierarchy of layers and their structural integrity. Prepare separate cut and engrave layers.

FABRICATION CONDITIONS

A5 portrait: 148 × 210 mm.
Grey cardboard: 1–2 mm.
Use millimetres, remove duplicate lines and avoid segments below 0.5 mm.
Complete the RapLab introduction and prepare for a 30-minute cutting slot.



Teaching references: Futurium, Menara Airport and the Córdoba Museum demonstrate depth and filtering in architectural surfaces.



Assignment example: three laser-cut layers and their assembled physical pattern.

Computational Design II · Assessment

Submission evidence connects geometry, design reasoning and material decisions

FS2025 · Assignment grading and tutor-supported exercises

SUBMISSION EVIDENCE

Blocks 1–3 require a poster and a short explanation, supported by a Grasshopper canvas and Rhino/Grasshopper files. Assignment 1 also asks for the Python script.

Block 4 replaces the poster with a photograph of the laser-cut model and an image of the cutting file, alongside the explanation and workflow files.

ASSESSMENT FOCUS

Completeness and adherence to the brief.

Clear visual composition and suitable technique use.

Geometric control and integration of computational methods with design intent.

Advanced computation where relevant, and an informed understanding of the architectural and fabrication concepts.

REFLECTIVE WRITING

Code: explain how geometry changes across iterations.

Analysis: explain the method and what it reveals.

Search: connect fitness criteria to the design outcomes.

Fabrication: justify geometry, material, thickness and layered tectonics.

TEACHING SUPPORT

Lecture recordings, assignment walkthroughs and tutorials support independent preparation. Flipped-classroom sessions provide direct tutor assistance.

Grading reads the final image together with the explanation and working files to assess how the technique supports the design decisions.

Computational Design III

Generative design and visualization introduce critical applications of AI

HS2025 · 052-0635-00L · ETH Zurich

HS25 DIGITAL THEORY & CULTURE SCHEDULE

Week #	Date	Block	Course	Submission
HS25	**12:00-13:30**			
38	18-Sep	**DESIGN SYNTHESIS**	Lecture 1	
39	25-Sep		Exercise Session Assignment Part I	**Part I due 1-Oct, 18.00**
40	2-Oct		Exercise Session Assignment Part II	
41	9-Oct		Exercise Session Assignment Part II	**Part II due 10-Oct, 18.00**
42	16-Oct		Course Forum	
43	23-Oct		*Seminar week*	
44	30-Oct	**COMMUNICATE SPACE**	Lecture 2	
45	6-Nov		Exercise Session Assignment Part I	
46	13-Nov		Exercise Session Assignment Part I	**Part I due 14-Nov, 18.00**
47	20-Nov		Exercise Session Assignment Part II	
48	27-Nov		Exercise Session Assignment Part II	**Part II due 28-Nov, 18.00**
49	4-Dec		Course Forum	

HS2025 teaching calendar: lectures, supported exercises, staged submissions and course forums for two thematic blocks.

BLOCK 1 DESIGN SYNTHESIS

Parametric modelling and AI-assisted coding in Grasshopper. Students examine how generative tools affect practice and authorship.

BLOCK 2 COMMUNICATE SPACE

Physically based rendering in Blender and AI-assisted rendering. Students connect computer graphics principles to architectural visual communication.

Computational Design III · Learning Goals

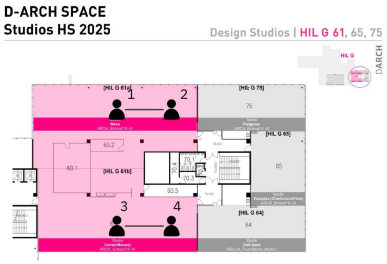
Technique, representation and critical reflection form a shared learning framework

HS2025 · Tutorials and exercise sessions

DIGITAL SKILLS

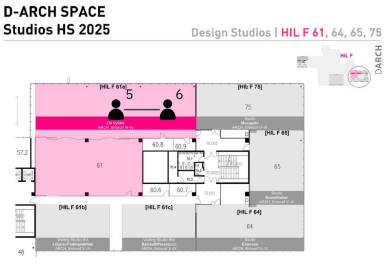
Implement and modify a generative workflow with and without AI. Document the process and parameters clearly.

Use digital rendering to communicate a stated architectural intent.

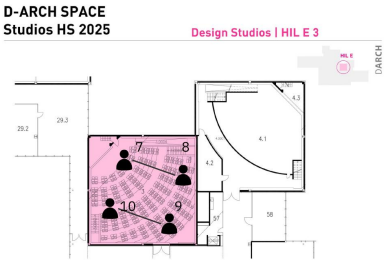


HIL G 61 (Mesa)
2 groups

HIL G 61 (Comte/Meuwly)
2 groups



HIL F 61 (De Vylder)
2 Groups



HIL E 3 (Kajima) and No Studio
4 Groups

HS2025 studio allocation supports tutor-led exercises across the HIL spaces.

COMPUTATIONAL UNDERSTANDING

Compare computational approaches and identify their architectural use cases.

Apply the working principles of computer graphics and AI-enabled visualization strategically.

CRITICAL REFLECTION

Evaluate authorship, bias and design intent. Discuss digital cultures from ethical and environmental perspectives.

Assess how visual representation influences perception, decisions and design processes at different project stages.

Working with the Existing

The HIL building scan becomes a library for architectural spatial collage

Computational Design IV · Block 3 · HIL Living Lab dataset

ASSIGNMENT

Reimagine one space or a relationship between adjacent spaces in the HIL building. Add, extract, move, remove and restyle scanned fragments to realize a clear spatial idea.



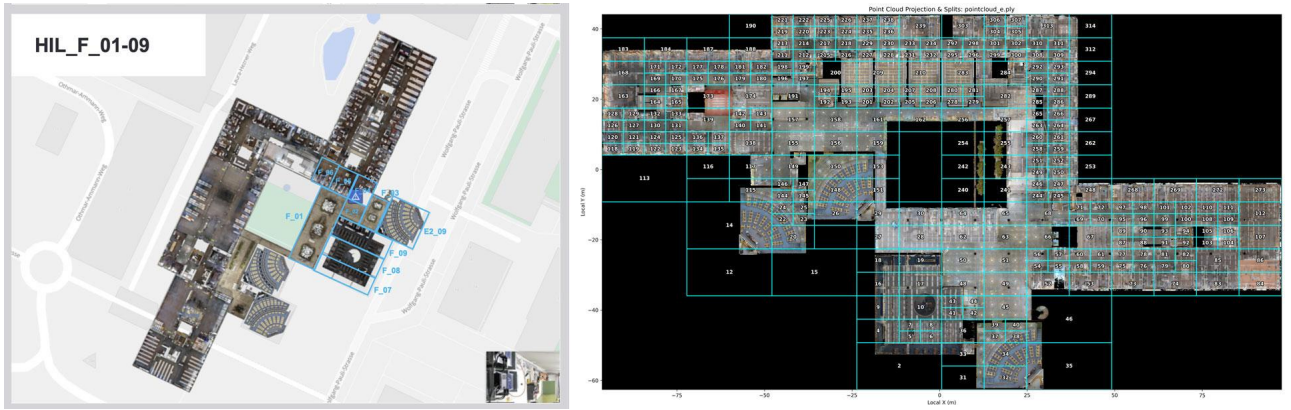
Teaching references position spatial collage between conceptual intervention, Matta-Clark's cuts and Chipperfield's Neues Museum renovation.

POINT-CLOUD WORKFLOW

Explore the web viewer, choose a spatial focus and import the selected data into Rhino or Blender.

Crop and place fragments as architectural parts. Use limited modelled geometry where it supports the concept.

Choose a visual strategy: realistic or abstract, dense or translucent, seamless or deliberately contrasting.



Shortlisted HIL spaces and the complete building scan offer different scales of spatial exploration.

Spatial Collage · Before & After

Matched viewpoints make the architectural intervention legible

Computational Design IV · Block 3 · Teaching example



HIL Alumni Lounge example: existing space and a proposed double-volume expansion. Roberto Innocenti and Nicola Piccioli-Cappelli.

REQUIRED EVIDENCE

Matched before/after images: 1024 × 1024 px.
Walkthrough GIF: 5–10 seconds.
Intent text: up to 100 words.
Reflection: up to 150 words.
A workflow screenshot shows fragments and camera setup.

ASSESSMENT

The visual intervention must express the written intent. The workflow and reflection explain why the chosen fragments, placement and material treatment create a distinct spatial character.

Hybrid Beasts · Assignment Brief

A continuous spatial element combines architectural functions and printing logic

Computational Design IV · Block 4 · Making Physical

ASSIGNMENT

Design a spatial element for a specific situation within the HIL building. Combine at least two structural, functional or aesthetic roles within a thick boundary or inhabitable surface.



3D-printing teaching references include IaaC's earth wall, Concrete Choreography, DBT Hollow Core printing, House Zero and HANNAH's Cores.

DESIGN AND FABRICATION

Use Rhino and Grasshopper to relate geometry to a continuous toolpath and G-code. Produce a 1:20 physical prototype that simulates large-scale printing.

A jury may select segments of up to three projects for a separate 1:1 fabrication opportunity.

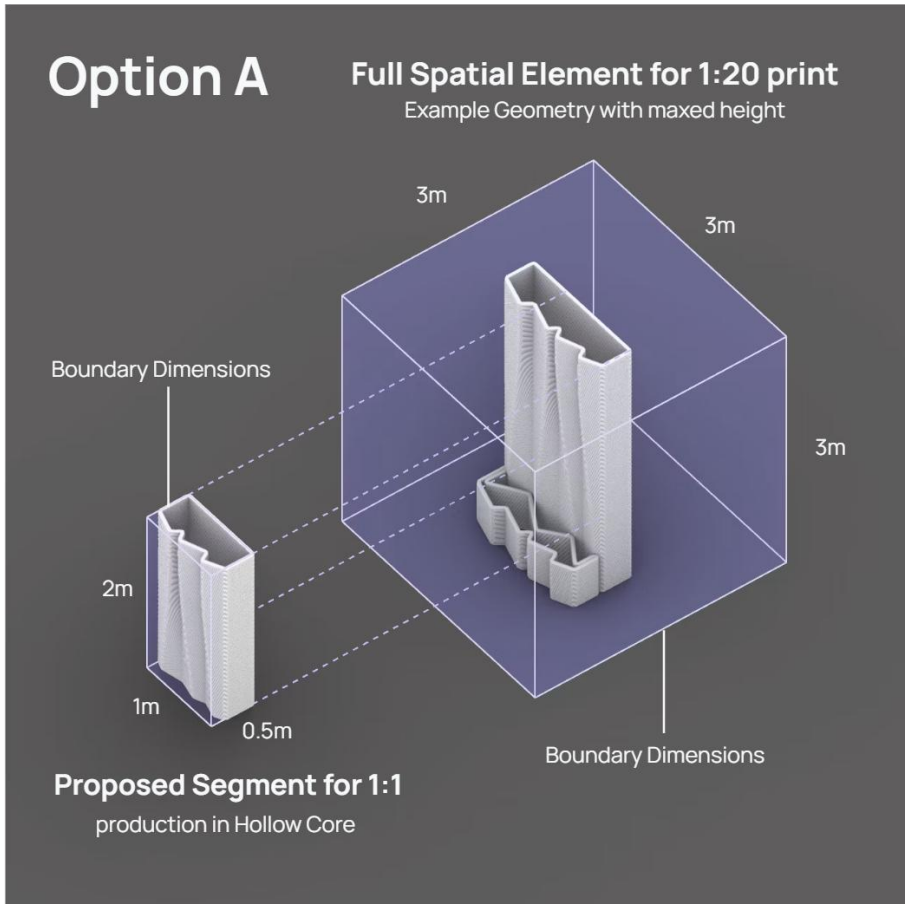


Architectural references: The Function of the Oblique and Sou Fujimoto's houses show surfaces supporting several modes of occupation.

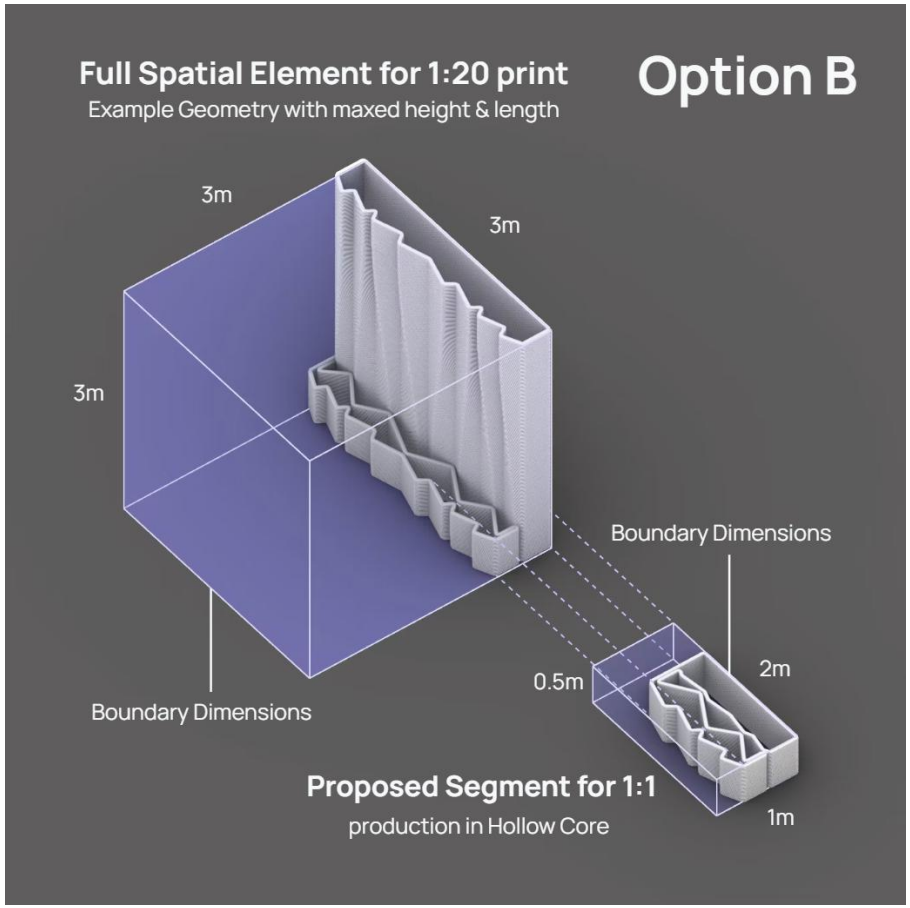
Hybrid Beasts · Design Constraints

Printing requirements guide the form and selection of a full-scale segment

Computational Design IV · Block 4 · Assignment diagrams



Option A: a full-height spatial element and a segment proposed for Hollow Core printing.



Option B: a full-length spatial element and a differently oriented 1:1 segment.

Full element: 3.0 × 3.0 × 3.0 m maximum.
Span the full height or one horizontal dimension.
Integrate at least two functions.

Proposed 1:1 segment: 2.0 × 1.0 × 0.5 m.
One continuous, uninterrupted toolpath.
Self-supporting geometry; ≤45° overhangs recommended.

Hybrid Beasts · Fabrication Workflow

Geometry, toolpath and G-code form a traceable design-to-production process

Computational Design IV · Block 4 · Rhino and Grasshopper

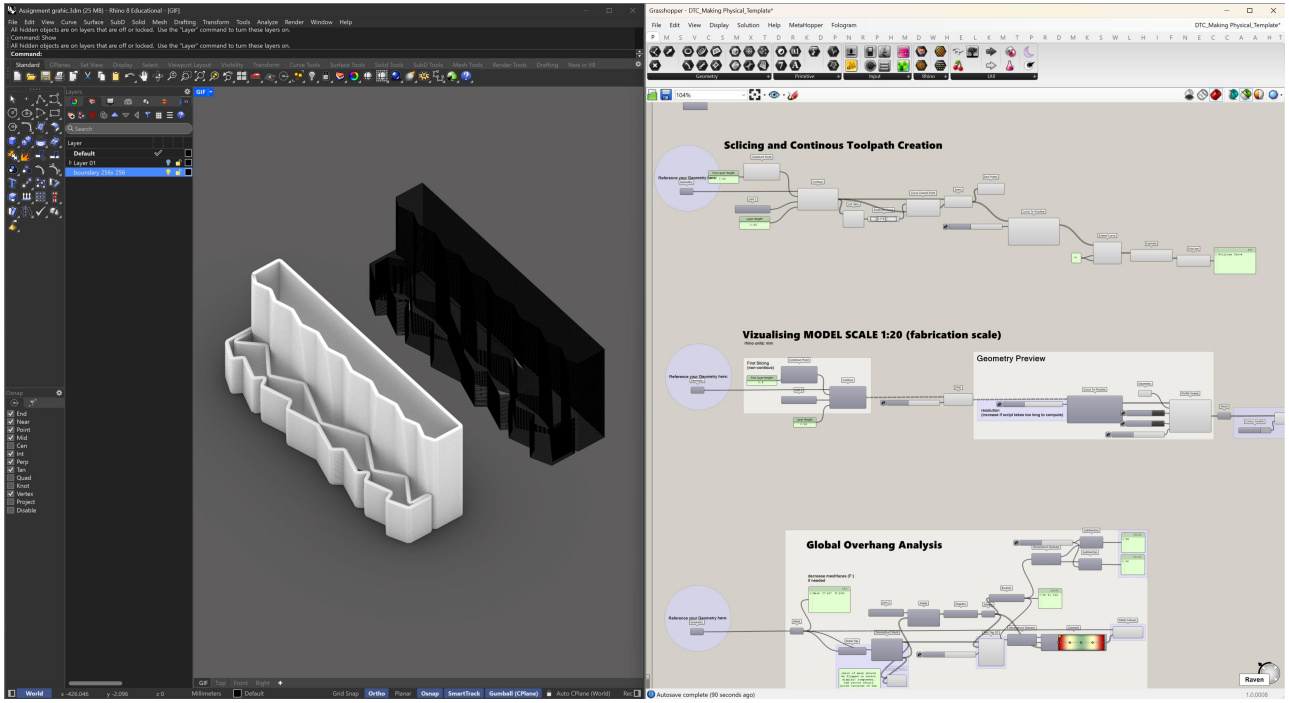
GEOMETRY AND PRINTABILITY

Model in millimetres at 1:20 scale. Avoid self-intersections and use turning radii compatible with extrusion width. Refine the architectural intent together with printability.

TOOLPATH AND DOCUMENTATION

Use the supplied slicing script to produce a single uninterrupted polyline. At 1:20 scale, the specified layer height is 0.56 mm and layer width is 2.0 mm.

Visualize the sequence with Lumbricus and export a layer-by-layer GIF.



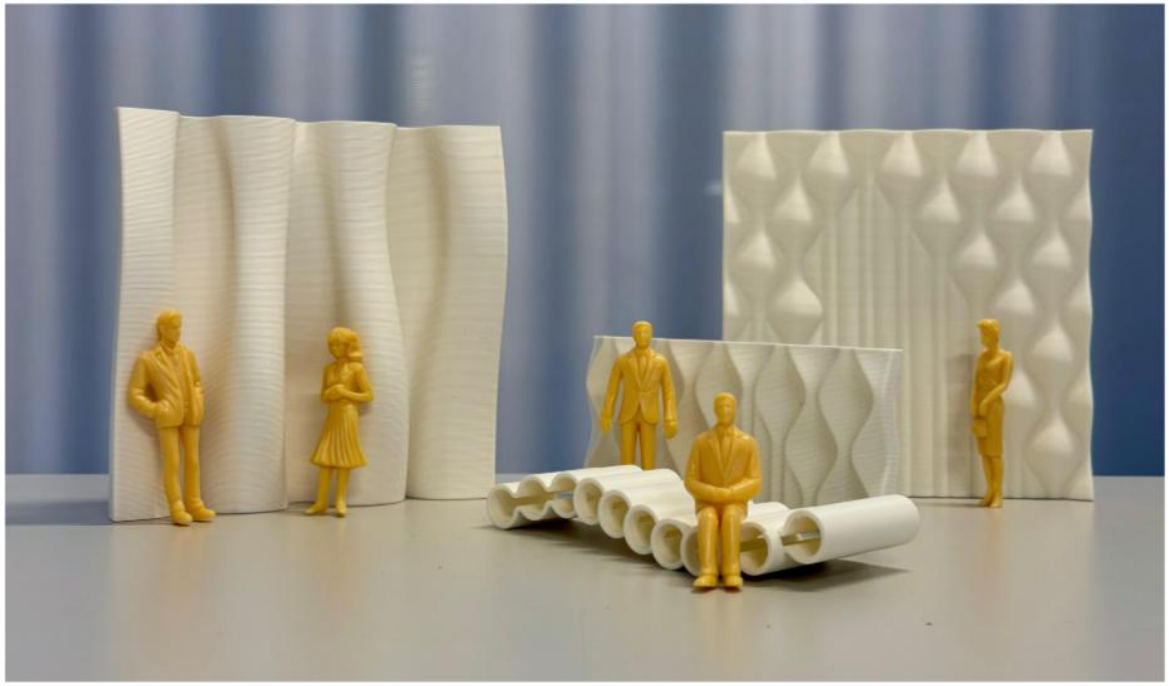
Teaching workflow example: a spatial element in Rhino and the supplied Grasshopper slicing, toolpath and printing controls.

Generate G-code with the supplied script, then fabricate the prototype on a Bambu A1 printer. Stage and photograph the result so its spatial affordances and fabrication logic remain clear.

Hybrid Beasts · Prototype Examples

Physical models make spatial affordances and printing logic visible

Computational Design IV · Block 4 · Teaching prototypes





1:20 teaching prototypes illustrate continuous printed surfaces with seating, enclosure and other affordances.

PHYSICAL EVIDENCE

The prototype tests whether the continuous geometry can be manufactured and whether its intended uses remain legible at model scale.

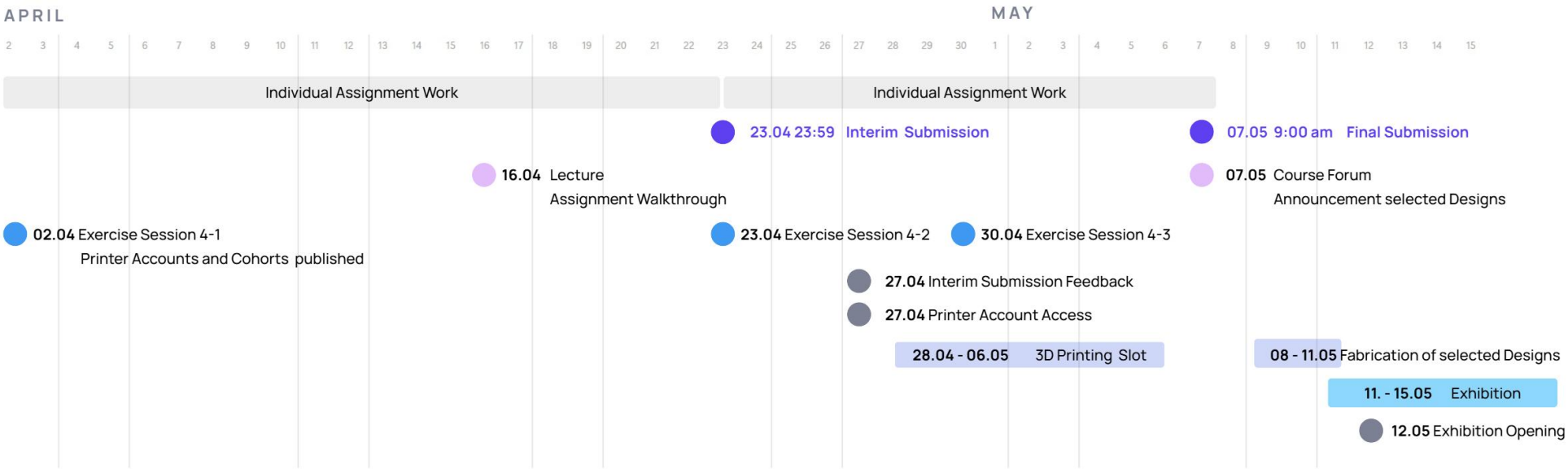
DOCUMENTATION

A staged photograph uses a neutral background and clear lighting. The final design graphic also identifies the segment proposed for possible 1:1 printing.

Hybrid Beasts · Teaching & Submission

An interim review supports revision before physical fabrication

Computational Design IV · Block 4 · FS2026



Assignment timetable coordinates exercise sessions, interim feedback, a shared printing window and final discussion.

TEACHING AND FABRICATION

Lecture: 16 April. Exercises: 2, 23 and 30 April.
Interim submission: 23 April, 23:59.
Printing: 28 April, 12:00 to 6 May, 23:59.
Final submission: 7 May, 09:00.

60 printers are organized in 20 shared accounts. Cohorts coordinate access to white matte or transparent PLA.

SUBMISSION EVIDENCE

Interim: print-sequence GIF, full-design image, geometry/toolpath file and a brief intent statement.

Final: physical prototype, staged photo, design/segment graphic, updated GIF, Rhino/Grasshopper files, intent and reflection texts.

Selection for 1:1 fabrication is separate from assignment grading.

Hybrid Beasts · Assessment

The rubric tests clarity and the synthesis of architectural intent with fabrication

Computational Design IV · Block 4 · Qualitative grading criteria

Criterion	Insufficient	Sufficient	Strong
Graphic clarity	The element, functions or chosen segment are unclear.	Images show the element, affordances and selected segment.	Precise modelling makes the element and functions immediately legible.
Text clarity	Intent, site, functions or the role of printing are unclear.	The text states the idea, site and intended functions.	A precise argument integrates site, affordances and fabrication.
Workflow clarity	Translation to a continuous toolpath and prototype is unclear.	The workflow shows geometry, toolpath and prototype.	The design integrates continuous printing, support-free geometry and prototyping.
Text–graphic	Images do not support the stated spatial intent.	Images make the stated intent and affordances visible.	Text and images strengthen the spatial argument together.
Text–workflow	The text does not address printing constraints.	The text explains key printing decisions in the design.	Fabrication constraints become architectural design drivers.
Synthesis	Concept, representation and fabrication do not relate.	Concept, representation and printing logic align.	Intent, representation and printing logic form a convincing whole.

These assignment-specific descriptors clarify the course grading criteria. The 1:1 fabrication selection does not affect the assignment grade.

Making computational design testable through teaching.

My teaching turns research systems into shared instruments for inquiry. Students test how data, constraints, fabrication, and material logic shape architectural decisions through workshops, modeling, prototyping, and critique.

Hang Zhang · Digital Building Technologies · ETH Zurich

hangzhang@ethz.ch

github.com/HangZhangZ ↗

2026